

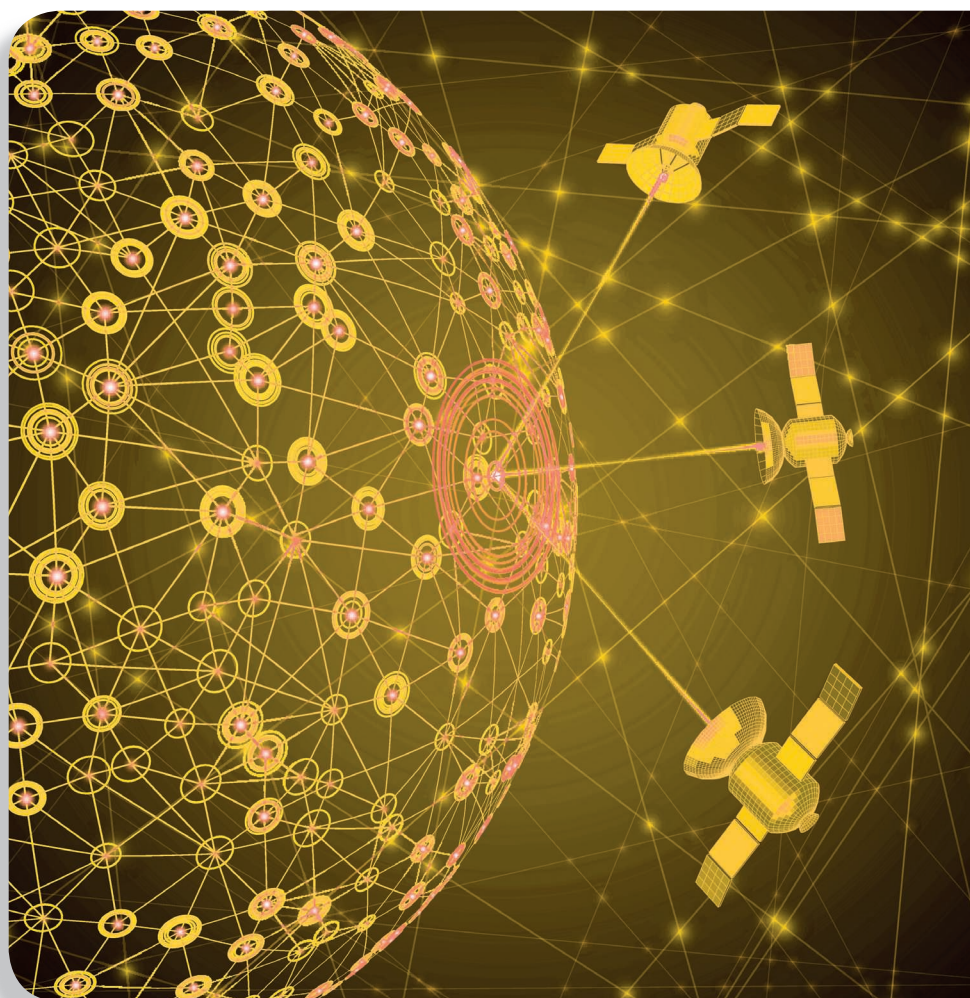
The field of satellite communications (SatCom) is enjoying renewed attention in the global telecommunications market, and high-throughput satellites, with their multiple spot beams, are critical components for delivering the rates that will be demanded in the future. In this article, we present the state of the art and the open research challenges in the area of high-throughput satellites, with a focus on signal processing approaches for efficient interference mitigation.

Our work sheds light on how some of the signal processing techniques that have been developed for wireless terrestrial communications can be applied in the context of satellites. All of the reviewed techniques are essential for empowering satellite systems to support the increasing demands of the upcoming generation of communication networks.

Introduction

SatCom systems, supported by their inherent wide coverage, are considered essential in satisfying data traffic ubiquity, which is expected to continue to increase over the coming decades. SatCom systems are a promising means to provide connectivity in unserved or underserved areas, regardless of the end user being fixed or on a moving platform on the ground, sea, or air (e.g., on a train, ship, or airplane) [1]. Also, satellites' ability to address wide geographic regions, even continents,

using a minimum amount of infrastructure on the ground, is extremely appealing. The ubiquitous coverage, together with the efficiency of SatCom's broadcast nature (i.e., its capacity to



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Signal Processing for High-Throughput Satellites

Challenges in new interference-limited scenarios

deliver the same content to a large number of users), has clearly been its main asset.

Note, however, that in recent years satellites have also developed broadband services [2], thanks to newly developed technologies that allow one satellite to manage hundreds of narrower beams with smaller coverage. These so-called spot beams can deliver different broadband data to different users. The new features improve the area data traffic they can support. Another appealing feature is that SatCom can be viewed as a green communication technology. This is because, once a satellite is in space, it has access to solar energy and can stay in orbit for up to 15 years with no additional energy source.

Thanks to these features, the SatCom ecosystem, on its own, can deliver valuable private- and public-sector results, such as resilient overlay communications in disaster relief, governmental services, traffic offloading and remote cellular backhaul provisioning (particularly for integrated access and backhauling, which are cornerstones of the upcoming mobile communications generations), supervisory control and data acquisition for telesupervision of industrial processes, and broadcast or multicast services. In both of the latter, multiple users receive the same information using the same radio resources. While *broadcasting* refers to the same information for the entire coverage area, *multicasting* denotes a beam or a set of beams. In this article, we try to stick to these meanings. However, these terms are often used interchangeably in the SatCom community, and *broadcast* may be sometimes used to encompass both.

While such a diversification of satellite-only services is foreseen to bear much fruit, the greatest benefits are envisaged as a result of integrating satellite and terrestrial communications in future communication generations. For instance, the roles and benefits of satellites in 5G [3] have begun to be studied in 3rd Generation Partnership Project (3GPP) Release 14, leading to the specific requirement of satellite-access support in TS22.261—*Service Requirements for Next Generation New Services and Markets; Stage 1*—recognizing the potential added value that satellite coverage brings as part of the mix of access technologies for 5G [4], [5]. In this context, potential new markets and emerging applications that are currently pursued by the satellite community include ubiquitous broadband access, commercial aeronautical and maritime services, machine-to-machine communications, and smart cache feeding.

In all of these applications, signal processing is challenged to satisfy the corresponding requirements in terms of spectrum and energy consumption. The increase in demand for these new satellite services and systems drives innovative approaches that are moving away from the traditional linear television broadcast (i.e., direct to the home). In addition, the development of new standards in the SatCom arena, some of them for satellite integration with the 5G New Radio standard, are helping to pave the way to meet the demand for faster and cheaper SatCom. By contrast, conventional commercial SatCom, which generally was custom-designed to meet the requirements of a specific mission, is losing importance. Notably, custom designs have been a limiting factor in reducing satellite cost and delivery time.

In parallel, an important trend of the past few years has been observed that relies on the vast potential of either high- or very

high-throughput satellites (H/VHTSs). Many operators are currently upgrading their constellations to deliver higher radio-frequency (RF) power, enhanced functionality, and higher frequency reuse with H/VHTS technology. With their flexibility, H/VHTSs play a key role in the paradigm shift that we noted previously. Their spectral efficiency (the number of bits per second per hertz) will be multiplied by two to three times over satellites without high throughput. In total, the expected aggregate high throughput will vary according to the application served and the satellite, but it is anticipated to be in the range of 25–60 gigabits/s. For VHTSs, the expected aggregate rate is within the terabit-per-second range.

This superior performance does not necessarily require higher bandwidth or increased payload weight. Therefore, a superior aggregate rate is provided at a lower cost per bit. As we present in this article, the H/VHTS approach leverages frequency reuse across multiple narrowly focused spot beams, thereby maximizing the available frequency band. However, high frequency reuse means higher levels of interference. This new trend poses interesting challenges regarding emerging interference-limited scenarios, and signal processing offers valuable tools to cope with them.

The purpose of this article is to provide an introduction to the new interference-limited paradigm that H/VHTS technology is facing, together with a thorough review of the recent signal processing approaches that address this new paradigm as well as new directions for research. In particular, we narrow our focus to fixed satellite services (FSSs) that are provided by geostationary satellites in the L, Ku, and Ka bands, where signal processing is needed to attain the promised high rates (at the Ku and Ka bands, these rates are on the order of a terabit per second). Notably, it is also in the geostationary orbit where H/VHTS originated, with well-established waveforms, coding, and modulators defined in the digital video broadcasting over satellite 2 extension (DVB-S2X) standard [6]. Both on-ground signal processing at the gateway (GW) and onboard signal processing at the satellite are addressed. Before that, let us comment, first, about the main features of SatCom systems and, second, about HTSs.

Features of SatCom systems

In past decades, SatCom systems exploited new techniques and technologies that had originally been implemented in terrestrial communications. For instance, in the mid-1980s, advanced analog-to-digital and digital-to-analog converters (ADCs and DACs, respectively) were used in delay-sensitive audio/voice applications, and satellite systems adapted them into the more complex digital signal processing techniques for delay-tolerant video broadcasting/multicasting. However, customization of terrestrial communication technologies to the peculiarities of SatCom systems is not straightforward because of the different satellite orbits, channels, system constraints, and processing. These peculiarities are commented upon next.

Today, there are approximately 1,300 fully operational communication satellites, and every type of orbit has an important role to play in the overall communication system. Geostationary Earth orbit (GEO), at 35,000 km, presents an

end-to-end propagation delay of 250 ms; therefore, it is suitable for the transmission of delay-tolerant data. Medium Earth orbit (MEO), at 10,000 km, introduces a typical delay of 90 ms; based on that, it can offer a compromise in latency and provide fiber-like data rates. Finally, low Earth orbit (LEO) is at between 350 and 1,200 km, and introduces short delays that range from 20 to 25 ms. In all of these cases, the satellite acts as a very particular wireless relaying node, the features of which lead to a communication system that cannot be always treated as a wireless terrestrial one [2]. The channel, communication protocols, and complexity constraints of the satellite system create a set of particularities, which are noted as follows:

- Because of the long distance to be covered from the on-ground station to the satellite, the SatCom link may introduce both a high round-trip delay and a strong path loss of hundreds of decibels. To counteract the latter, satellites are equipped with high-power amplifiers (HPAs) that may operate close to saturation and create intermodulation and nonlinear impairments (in much the same way as audio amplification).
- Usually, one HPA controls one antenna element (called a *feed*) on the satellite. To ensure that there is no distortion, it is critical to satisfy the power limitations on each feed, not only on the average total power that the satellite transmits.
- SatCom traverses about 20 km of atmosphere and experiences high molecular absorption, which is even higher in the presence of rain and clouds, particularly for frequencies above 10 GHz. Therefore, satellite links have been conventionally designed based on thermal noise limitations and on the link budget analysis that have to consider large protection margins for additional losses (e.g., rain attenuation).
- In the nongeostationary orbits (i.e., MEO and LEO), there are significant time-domain variations of the channel due to the relative movement of the satellites with respect to the ground station. Variations in Doppler can exceed the range $[-200, 200]$ kHz over the pass of the satellite (e.g., 45 min for MEO).
- Because of the long distance and carrier frequencies, the satellite antenna feeds are generally seen as a point in the far field, thus precluding the use of conventional spatial diversity schemes on board the satellite. Also, because of the absence of scatterers near the satellite (i.e., there are no objects in space that create multiple paths) and the strong path loss (i.e., it is a long-distance communication), the presence of a line-of-sight component, which focuses all of the transmitted power and is not blocked or shadowed, is required. This contrasts with terrestrial cellular communications, where the presence of a line of sight is not critical. On the positive side, because of the lack of rich scatter, the SatCom experiences higher cross-polarization isolation than terrestrial communication networks.
- The processing complexity aboard the satellite is limited and largely unexplored, as it is highly correlated with its power consumption, mass, and, ultimately, with the cost of the system.
- The received signal-to-noise ratio (SNR) is very low, and, therefore, the user terminal (UT) must have high sensitivity,

good receiver antenna gains, and good tracking capabilities to steer the UT beam, such that it continuously points to the satellite.

- The practical challenges of a satellite system require solutions different from the ones used in terrestrial wireless communications. An important one is the specific satellite multiuser protocol framing that is defined in the current broadband and mobile interactive standards, DVB-S2X and BGAN, respectively. In these protocols, to overcome the satellite channel noise, channel codes are long and, therefore, must take into account data from multiple users. This fact creates a multicast transmission, because the same information has to be decoded by a group of users. Multicast transmission requires specific signal processing techniques, as this article explains.
- Finally, satellite solutions are generally characterized by relatively long development and in-orbit validation phases before becoming operative. This is different from terrestrial solutions, where it is easier to test new technologies in situ without incurring excessive deployment costs.

As we have commented, this article focuses on the recent H/VHTS technologies because of their potential to improve SatCom systems in both performance and cost. Next, we describe these technologies in detail.

HTSs: A new interference-limited paradigm

In contrast to monobeam satellites, HTSs partition the service area into multispot beams; this allows a higher aggregate throughput and more service flexibility to satisfy the heterogeneous demand. The system architecture is shown in Figure 1 and consists of a GW, a satellite, and multiple UTs. The GW is connected to the core network and serves a set of satellite users geographically far away. The links from the GW to the satellite and from the satellite to the UT are known as the *feeder link* and the *user link*, respectively. In the usual star configuration illustrated in Figure 1, the feeder link presents high directivity and gain. As this link has an SNR considerably higher than the one in the user link, it is generally assumed to be noiseless and perfectly calibrated against channel power variations. Also, depending on the direction of the communication, the link is known as a *forward link* when it originates at the GW to the UT and as a *reverse link* when it travels from the UT to the GW. These links combine to create the feeder forward, user forward, feeder return, and user return links, each of which usually works in different frequency bands.

The frequency selection is driven by considering different aspects, among them coverage and beam size, atmospheric conditions in the served region, and availability of a robust ecosystem of ground equipment technologies. In SatCom, the signals have to cross the atmosphere and use high frequencies. Otherwise, the signals would be reflected by the atmosphere and could not penetrate to the satellite. For instance, current-generation GEO HTSs typically use the Ka band, which is less congested than the C and Ku bands. For FSSs, this refers to the exclusive satellite band from 19.7 to 21.2 GHz for the forward link and from 29.5 to 31 GHz for the reverse link.

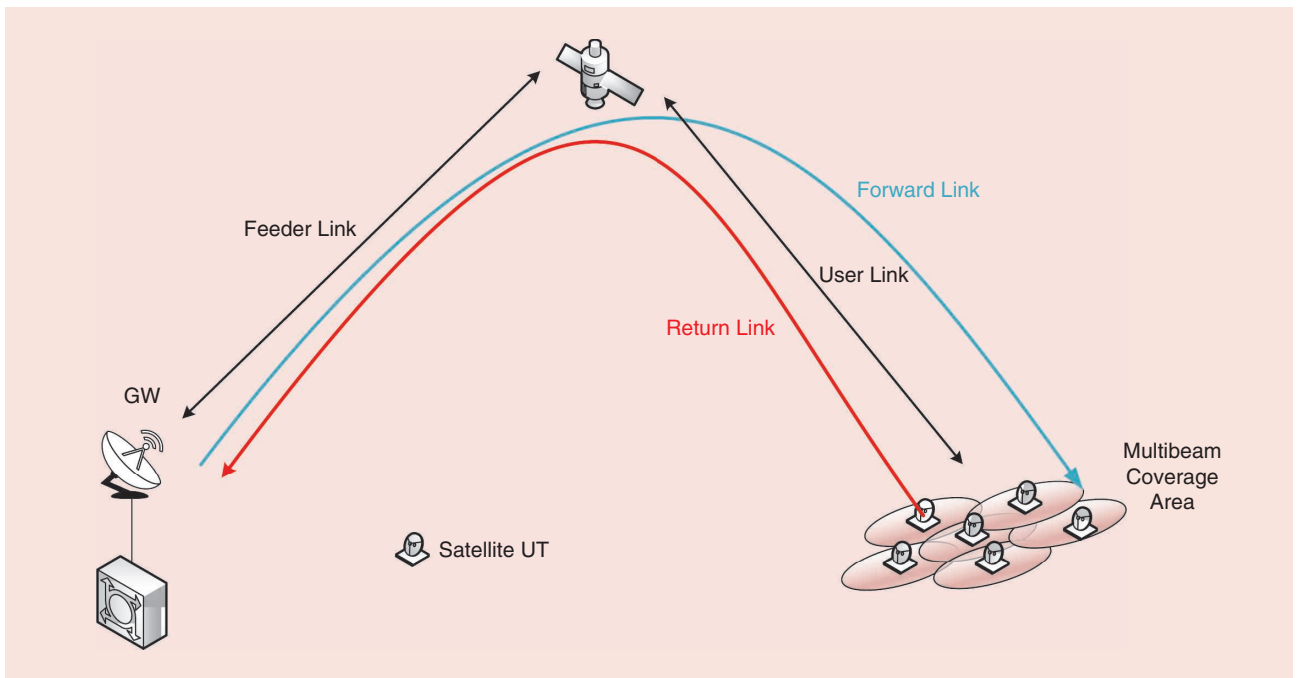


FIGURE 1. The scheme of a multibeam satellite system. The forward link goes from the GW to the UTs via the satellite (blue line). The return link travels from the UTs to the GW via the satellite (red line). The feeder link connects the GW with the satellite (bidirectionally), whereas the user link connects the UTs with the satellite (bidirectionally).

Mobile satellite services generally use lower frequencies, such as the L band (i.e., from 1.5 to 2.5 GHz), because of their lower attenuation, which enables lower antenna gains at the mobile UT. Note, however, that recently the Ka band is also being considered to provide in-flight and maritime connectivity. In all of these cases, the frequencies transmitted by the satellite occupy the lower band. The satellite is a lightweight device that cannot support high-power transmission units. Therefore, it transmits at a lower frequency (the higher the frequency, the higher the transmission power to accommodate losses) as compared to the GW, which can afford to use very high-power transmission.

The HTSs that are currently operative (e.g., *Viasat-2* and *SES-12*) provide aggregate data rates of more than 100 gigabits/s. These HTS systems use the Ku and Ka bands in both the feeder and user link and serve as many as 200 beams in the user link. For their part, H/VHTS systems (e.g., *Viasat-3*) aim at achieving data rates in the range of a terabit per second and, because of that, they need higher frequencies in the Q band (30–50 GHz), V band (50–75 GHz), and W band (75–110 GHz) to serve as many as 3,000 beams in the user link. For these reasons, advanced signal processing techniques are required to reduce the interference among such a plethora of beams, facilitate adaptive coverage, dynamically optimize the traffic, and share the spectrum with terrestrial services, among other functions. Flexibility in the resource allocation per beam can significantly improve the quality of service (QoS) and bring down the incurred cost of the H/VHTS system per transmitted bit.

Figure 2 shows an example of classical linguistic beam wide coverage. In contrast, multispot beams allow tessellation of the coverage into much smaller footprints, thus enabling frequency

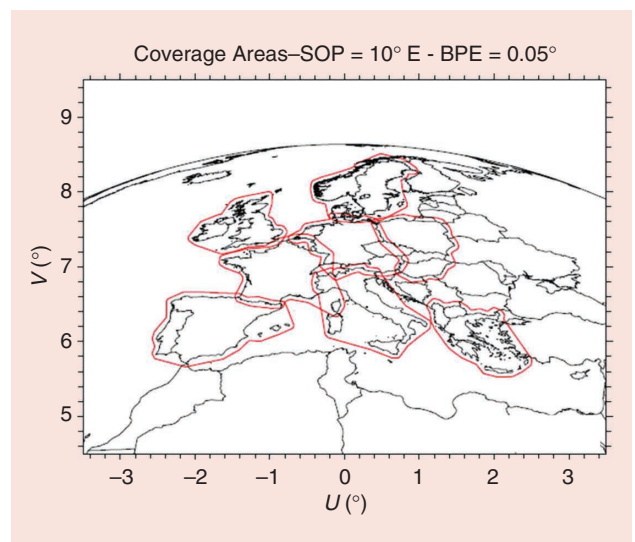


FIGURE 2. A broadcasting satellite with eight linguistic beams in the Ka band. (Source: © European Telecommunications Standards Institute, 2015; further use, modification, copy and/or distribution are strictly prohibited.)

reuse within the geographical area covered by one linguistic beam. As a consequence, the per-user bandwidth assignment and the aggregate throughput can potentially increase in H/VHTS. Multispot beams enable broadband data services in addition to the traditional broadcast services offered by the linguistic beams.

Figure 3 shows an example of the footprints of a four-color reuse scheme, where a total bandwidth of 500 MHz is allocated

to the user link at the Ka band. This bandwidth is divided into two sub-bands that, when combined with two orthogonal polarizations, generate the so-called four-color beam pattern across the coverage area. In the Ku and Ka bands, orthogonal polarizations maintain very low cross-polarization and, because of that, they can be used as if they were different frequencies. Within each beam, multiple users can be served with the aid of an orthogonal access scheme. Currently, with the common frequency reuse of four colors, the interference power among beams is in the range from 14 to 34 dB below the carrier signal.

With the aim of lowering the cost per transmitted bit and increasing the spectral efficiency or the available system bandwidth, new systems aim at more aggressively reusing the available spectrum among the spot beams. This aggressive frequency reuse may not be applied to all of the satellite beams but only to those experiencing a higher traffic demand; thus, the performance indicator in this case is the area traffic capacity. Nevertheless, increasing the frequency reuse leads to a further increase of intrasystem interference among the cochannel beams, which shifts the classical noise-limited link budget analysis toward an interference-dominated situation. The sidelobes of the beam radiation patterns create interference leakage among beams, and the carrier-to-interference ratio (CIR) can be severely degraded.

To successfully implement high frequency reuse, interference management has to be implemented at the GW, the satellite, the UT, or some combination of these. It follows that the CIR depends mostly on the position of the UT, the crossover level, and the antenna radiation pattern. Hence, the most favorable case corresponds to the situation in which the UT is in the center of the beam, while the worst case is when the UT is located in the beam-edge area. We note that, for a frequency-reuse pattern equal to one (i.e., $f_r = 1$), the median of the CIR is around 0 dB and ranges from -5 to 15 dB; for $f_r = 2$, it is 8 dB and spans from 0 to 25 dB; for $f_r = 3$, it is 25 dB and ranges from 10 to 30 dB; and, for $f_r = 4$, it is 30 dB and reaches from 10 to

40 dB. In any case, the interference power that comes from the high frequency reuse adds to that originating from the nonlinear distortion of the HPA. Unfortunately, the traditional approach to diminish interference by using power control is insufficient, and, hence, novel signal processing alternatives that exploit the structure of the cochannel interference are needed.

We note that the final performance of the H/VHTS system depends not only on the capabilities of the applied signal processing but also on many system choices. Complex design tradeoffs and practical aspects need to be respected, as detailed in [7]. For example, if hundreds of beams are available in the system, high frequency-reuse schemes can stress the payload resources of the satellite in terms of mass, power, and thermal dissipation. Another important consequence of increasing the frequency reuse is that the bandwidth of the feeder link should increase accordingly. As this is not straightforward, different alternatives should be studied, such as employing multiple GWs in the feeder link. Provisioning signal processing on board the satellite, i.e., onboard processing (OBP), can provide additional degrees of freedom toward mitigating the interference and enabling seamless multiple GW operation; the choice of OBP depends on mission limitations and commercial viability.

Finally, it is important to note that H/VHTS systems require the most advanced transmission standards. Currently, DVB-S2X in [6] is the standard for both forward broadcast and broadband satellite networks. We can say that such standards have been designed to best fit the features of the GEO satellite channel (i.e., its delay and transmission power limitations). Using high-efficiency modulation and coding schemes (MODCODs) up to 256 amplitude phase-shift keying, combined with advanced interference management techniques, allows for aggressive and flexible frequency reuse. DVB-S2X incorporates the novel superframing structure that enables the use of signal processing techniques that have never been employed before in the satellite context, such as precoding and multiuser detection at the UT. Among other things, this standard introduces orthogonal Walsh-Hadamard sequences as reference/training sequences, allowing simultaneous estimation of the channel state information of multiple beams.

The superframe concept is designed to maximize the efficiency of the channel-coding scheme by encapsulating the information intended for several UTs using the same MODCOD. The novel superframe specification has opened a new scope of research related to advancing interference management techniques for aggressive frequency-reuse scenarios. Specifically, it is a multicast transmission because, in the framing structure of DVB-S2X, the same data are transmitted to multiple receivers. This precludes the calculation of a precoding matrix on a receiver-by-receiver basis. Differently, a multicast precoder has to be designed for a group of receivers. Further details on the DVB-S2X standard that have a beneficial impact on precoding and multiuser detection can be found in [6, Annex E]. Although DVB-S2X is the most popular satellite standard today because of its design for broadband and broadcast communications, the other most-used standard is BGAN. This is for mobile interactive communications, and

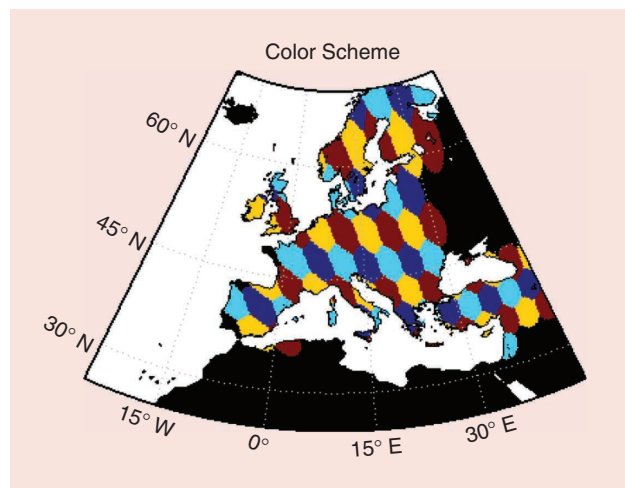


FIGURE 3. The user frequency plans for the scenario with 71 beams and a frequency-reuse pattern equal to four. (Source: © European Telecommunications Standards Institute, 2015; further use, modification, copy and/or distribution are strictly prohibited.)

it also uses a similar superframe concept. Next, we address spatial precoding techniques, which are supported by these standards to mitigate the interbeam interference.

Precoding in multibeam satellite systems

Architecture and communication peculiarities

With the aim of increasing the offered data rates of a given satellite, both operators and manufacturers are investigating a variety of alternatives. One main approach is to consider SatCom links at extremely high frequencies, such as the W band. However, large investments are required for implementing the communication subsystems in these bands, because no previous commercial SatCom systems have been operating at these frequencies. In addition, new challenges with regard to channel impairment appear. As a result, spectral-efficient alternatives that exploit the current frequency bands are receiving much attention.

This is the case with precoding techniques that allow a high frequency-reuse factor among the different beams in the user link. With the aid of precoding, a satellite UT can obtain a sufficiently large signal-to-interference-plus-noise ratio (SINR), even though the carrier bandwidth is reused by adjacent beams. This is

because, to maintain a certain SINR value, the precoder uses the channel knowledge to mitigate the interference toward the UTs. Resorting to the system architecture that is schematically depicted in Figure 1, the precoding matrix is, in general, computed at the satellite GW. After that, the beam signals are precoded and transmitted through the feeder link using a frequency-division multiplexing scheme. Then, the satellite payload performs a frequency shift and routes the resulting radio signal over an antenna array that transmits the precoded data over a larger geographical area that is served by the multiple beams in the user link. The block diagram is summarized in Figure 4.

Low-complexity linear precoding techniques are of great interest, and this is a problem that deserves further attention and research. Indeed, the computational complexity that is required to implement multibeam satellite precoding techniques gains even a larger importance compared to cellular systems, as the dimensions of multibeam satellite systems are extremely high. For instance, the forthcoming *Viasat-3* system is expected to utilize nearly 1,000 beams per color to serve the coverage area presented in Figure 5. As a result, the on-ground equipment should be prepared to update a precoding matrix to cope with 1,000 users on a per-frame basis. This complicates

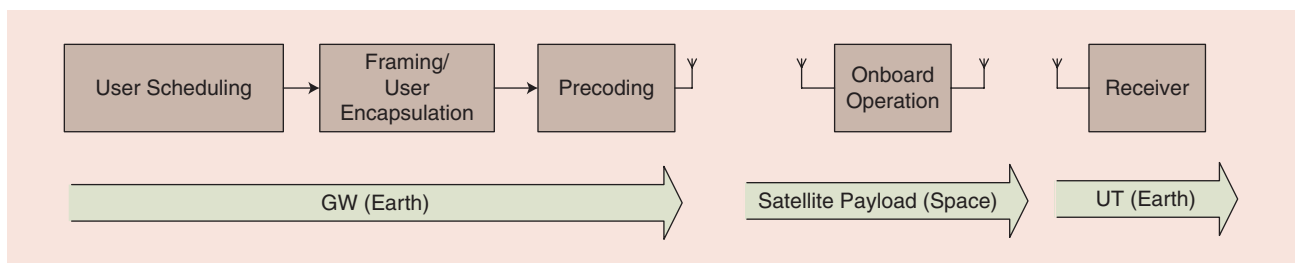


FIGURE 4. A block diagram with the multibeam satellite precoding at the GW.

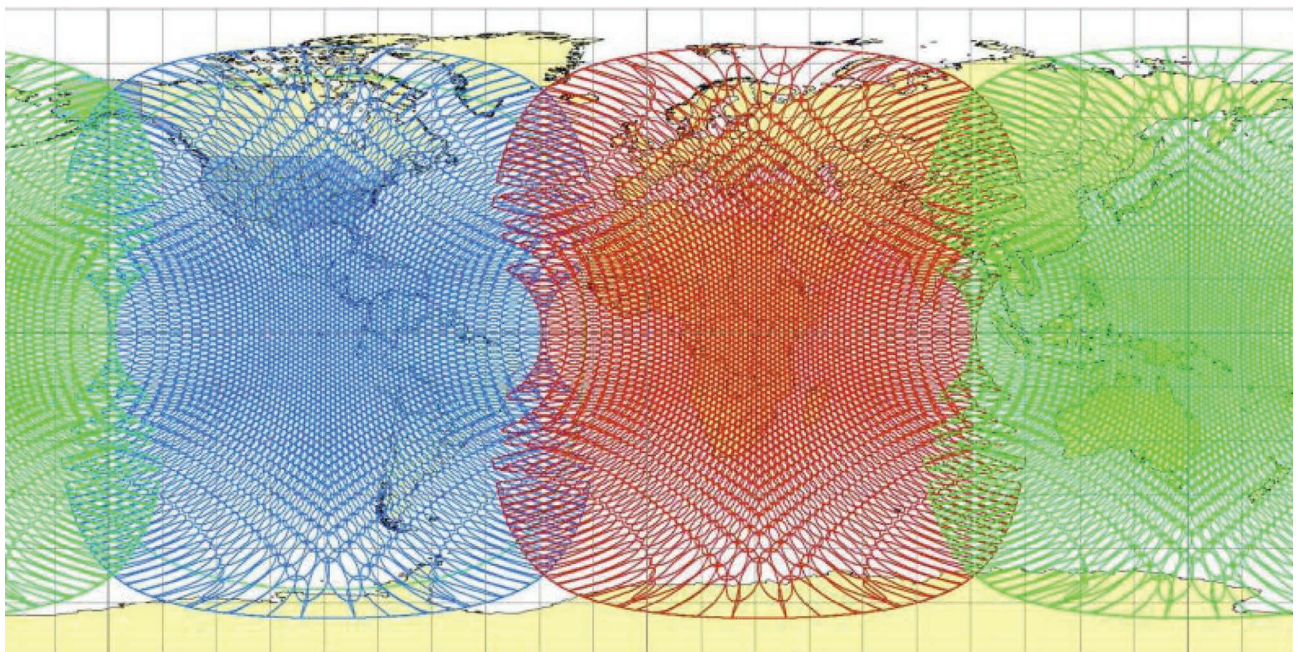


FIGURE 5. *Viasat-3* beam pattern footprints. Each of the colors corresponds to three different satellite coverage areas. There are 1,000 spots per color. (Image courtesy of Viasat.)

the precoding implementation because of the extremely large size of the precoding matrix that must be calculated (i.e., a $1,000 \times 1,000$ matrix). Further details are explained in this section after the introduction of the mathematical system model (i.e., the “Precoding Techniques” section).

To create all of these beams, at least an equal number of antenna elements is needed on board the satellite. Bear in mind that, with these large dimensions, we could draw an analogy with a massive multiple-input, multiple-output (MIMO) system [8]. However, the actual multibeam satellite performance does not scale with the number of antennas and UTs as terrestrial systems do. The main reasons can be summarized as follows:

- *The multibeam satellite system is generally user overloaded:* Current satellite payloads are equipped with N antennas (generally called *feed elements*) that have to serve a total number of UTs much larger than N . This clearly contrasts with massive MIMO, where the number of base station antennas grows to be larger than that of the UTs. Therefore, as N grows in the massive MIMO framework, the angular resolution of the antenna array increases so users can be separated more effectively by the appropriate antenna weights, which does not happen in H/VHTS SatCom.
- *The cochannel interference power does not decrease as the number of beams increases:* The favorable propagation in massive MIMO mentioned in [9] does not occur in multibeam satellite systems. That is, in a scattered terrestrial channel environment, the off-diagonal elements of the channel covariance matrix tend to zero as the number of antennas grows, leading to an ideal interference-free scenario. In other words, the fast-fading channel from the different users becomes almost orthogonal as the number of antennas in each base station grows large while keeping the number of UTs fixed. On the contrary, because of the low scatter in the GEO satellite channel (i.e., the line-of-sight channel), there is always strong cochannel interference among the beams.
- *There is no pilot contamination:* Massive MIMO in multicell scenarios entails difficulties in the channel estimation operation, as base stations or users located in adjacent cells might inject interference into the estimation process in the downlink or in the uplink, respectively. This is because orthogonal pilot sequences have to be reused from cell to cell, as in massive MIMO systems each cell has a high number of antennas and users active in the same time/frequency resource. Ideally, for time-division multiplexing, one orthogonal sequence per transmitting antenna would be needed for each base station in the downlink and one per user transmitting antenna in the uplink, and this is not possible.

Note that, in frequency-division multiplexing, the users send their corresponding estimated channel to the base station. If this is done simultaneously by a large number of users, signal contamination may also arise. The multibeam satellite case is not the usual case, since, in the downlink, each UT has to estimate the channel from each beam, and, therefore, only one pilot signal per beam is needed. Because of the high directivity of the beams, only the adja-

cent ones are those that create a relevant interference. This is a limited number of beams, and, therefore, the pilot signals of adjacent beams can be orthogonal (i.e., note that DVB-S2X considers a set of 32 Walsh–Hadamard sequences that can be used as simultaneously sent pilots for the estimation of up to 32 channels in the downlink). For the same reason, in the uplink, either in frequency- or time-division multiplexing, the number of simultaneous transmitted signals for channel state information acquisition purposes is limited. Also, the satellite channel is, in general, not frequency selective and preserves the orthogonality at the UT. Pilot contamination could appear only in the case of having multiple satellites with independent processing of their signals.

In any case, although H/VHTS cannot be considered a massive MIMO system, it could benefit in the future from technology advances in terrestrial massive MIMO. This can be the case, for instance, with the use of larger antenna apertures to reduce the size of the spot beams to improve capacity and with capability of flexible formation of beams. This would contrast with the actual spot beams with fixed footprints (e.g., Figure 3).

Another interesting aspect is that precoding in multibeam satellite systems presents certain similarities with respect to the cloud radio-access network (C-RAN) architecture (e.g., see [10]). Referring to the description in Figure 6, in both C-RAN and multibeam satellite systems, the baseband processing is performed by a centralized entity termed in the C-RAN the *baseband unit (BBU)*. In the satellite, the feeder link plays the role of the optical fiber links from the BBU to the remote antenna. Therefore, we can say that a multibeam satellite is similar to a C-RAN cellular architecture with a wireless fronthaul in the sense that the baseband processing and the RF elements are placed in separated locations. However, in contrast to the C-RAN, multibeam satellites do not generally perform a digitalization of the baseband signal. In other words, they behave as an amplify-and-forward relay.

In light of the previous discussion, current multibeam satellite systems do not present the same peculiarities as a C-RAN. There exist substantial differences between the two approaches, mainly in terms of implementation details that reflect the nature of the underlying systems. Nevertheless, the progress on signal processing on board the satellite, and on the softwarization of the satellite networks, makes the future convergence of both systems almost inevitable. Thus, we identify C-RAN architectures that include the satellite component as a new direction worth investigating. Next, we introduce the system model to understand the peculiarities of the signal processing precoding for HTSs. The challenge is to cope with a frequency-reuse pattern equal to one.

System model

Consider a multibeam satellite system where each UT has only one antenna and the satellite is equipped with N antennas, which are combined to generate a beam radiation pattern composed of K fixed beams [11]. (In this section, we use the term *beam* when we refer to a satellite transmission of N_u UTs over

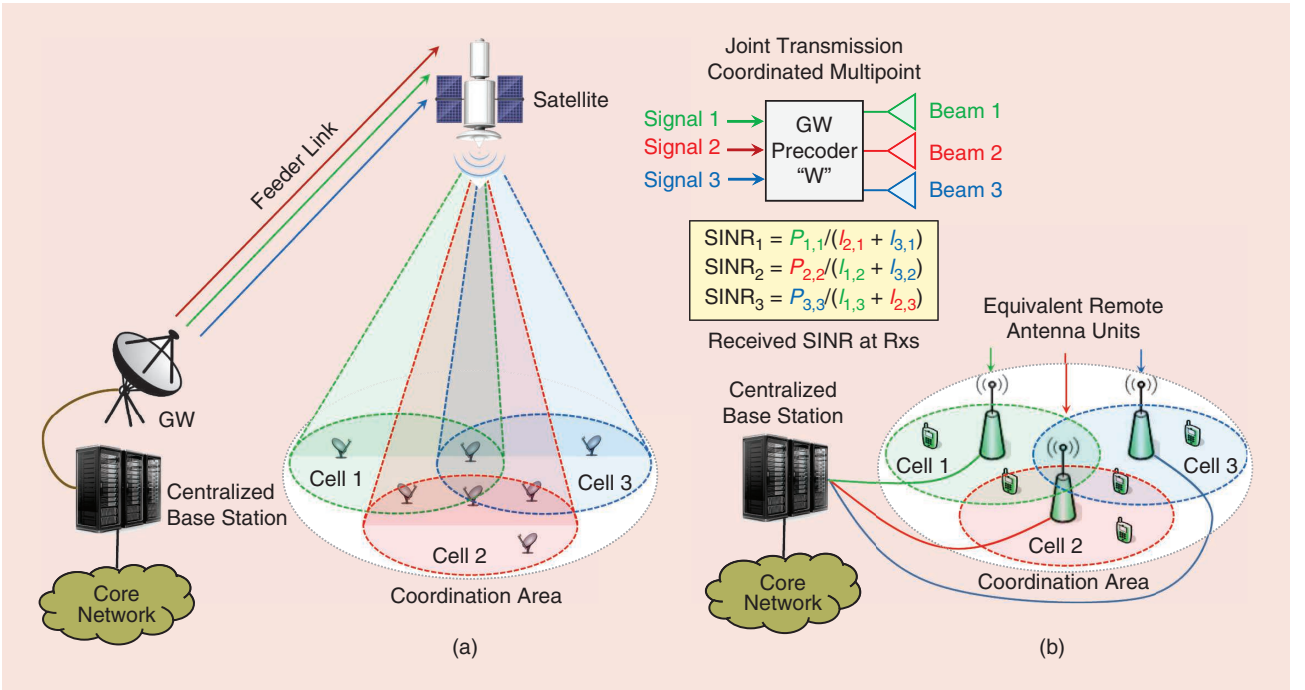


FIGURE 6. A multibeam satellite precoding system versus a C-RAN cellular system: (a) a multibeam satellite system and (b) an equivalent C-RAN system. Rxs: receivers.

a certain coverage area. Even though the term might be confusing for some readers and its original meaning *fuzzy* in systems with precoding, we use it because it is generally employed by the satellite engineering community.) For each beam, we assume that a total number of N_u users is simultaneously served per beam (i.e., the total number of served users by the satellite is KN_u).

Consider that all beams radiate in the same frequency band (i.e., $f_r = 1$). Single-carrier modulated signals are transmitted, which is the most usual case in SatCom because of the degradation that the HPA's nonlinearities create on multicarrier signals (e.g., DVB-S2X uses single-carrier modulations). The discrete equivalent baseband model for the received signal at the i th UT of each of the K beams is written as

$$\mathbf{y}^{[i]} = \mathbf{H}^{[i]} \mathbf{D}(\mathbf{x} + \mathbf{z}) + \mathbf{n}^{[i]}, \quad i = 1, \dots, N_u, \quad (1)$$

where vector $\mathbf{y}^{[i]} \in \mathbb{C}^{K \times 1}$ is the vector containing the received signals of the i th UT (i.e., the value $[\mathbf{y}^{[i]}]_k$ refers to the received signal of the i th UT at the k th beam), whereas vector $\mathbf{n}^{[i]} \in \mathbb{C}^{K \times 1}$ contains the noise terms of each i th UT. The entries of $\mathbf{n}^{[i]}$ are assumed to be independent and Gaussian distributed, with zero mean and variance equal to σ_n^2 (i.e., $E[\mathbf{n}^{[i]} \mathbf{n}^{[i]H}] = \sigma_n^2 \mathbf{I}_K$, $i = 1, \dots, N_u$). Vector $\mathbf{x} \in \mathbb{C}^{N \times 1}$ contains all of the transmitted signals (i.e., one per transmitting feed). The term $\mathbf{z} \in \mathbb{C}^{N \times 1}$ corresponds to the noise term of the feeder link transmission, assumed to be Gaussian, with zero mean and with variance equal to σ_f^2 .

Matrix $\mathbf{H}^{[i]} \in \mathbb{C}^{K \times N}$ is the channel matrix, whose k th row, denoted by $\mathbf{h}_k^{[i]}$, is the channel vector of the i th user located at the k th beam. This vector contains the channel coefficients from each antenna element $n = 1 \dots N$ to the user. To clarify this sys-

tem model, Figure 7 describes how $\mathbf{H}^{[i]}$ is constructed. As can be observed in the example with $K = 3$ and $N_u = 2$, the different rows of these channel matrices $\mathbf{H}^{[1]} \in \mathbb{C}^{K \times N}$ and $\mathbf{H}^{[2]} \in \mathbb{C}^{K \times N}$ are formed by UT channel vectors located at each of the beams.

Note that this is a flat-fading channel, which is the realistic assumption in the use cases for the Ku and Ka bands with the 500-MHz user link bandwidth, even for mobile satellite systems [12]. This is due to the high directivity of the beams, which eliminates any possible multipath. Matrix $\mathbf{D} \in \mathbb{R}^{N \times N}$ consolidates the feeder link and payload operation. Note that, in this section, we do not consider the nonlinearity onboard operation effects, which are analyzed in the following section. We write matrix $\mathbf{D}$ as follows:

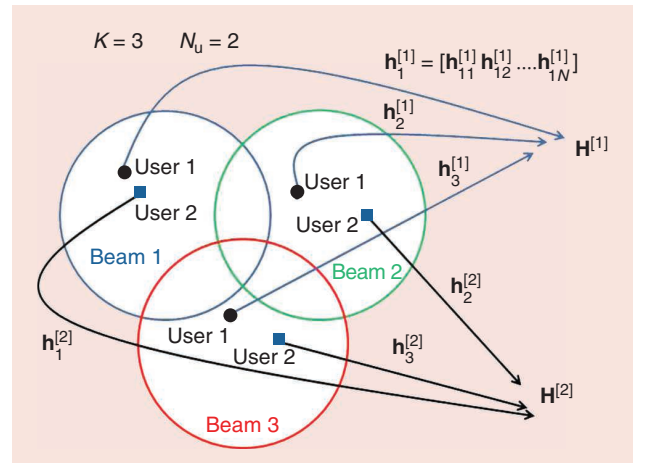


FIGURE 7. A system model example with $K = 3$ and $N_u = 2$. The channel matrices $\mathbf{H}^{[1]}$ and $\mathbf{H}^{[2]}$ are formed by the UTs and represented by squares and circles, respectively.

$$\mathbf{D} = \text{diag}(m_1, \dots, m_n, \dots, m_N), \quad (2)$$

where m_n denotes the resulting amplitude variation of the n th feed signal. Assuming perfect onboard automatic gain control mechanisms, we can establish that $m_n = 1$ for all $n = 1, \dots, N$, rendering $\mathbf{D}$ to be an identity matrix.

Feeder links are supported by parabolic antennas 7 to 9 m in diameter at the GW and a high-gain antenna at the payload. The feed signals are multiplexed in frequency over the feeder link's available bandwidth. In this context, it is always the case that the feeder link's SNR, which is defined as

$$\text{SNR}_{\text{feeder}} = \frac{E[|\mathbf{x}|^2]}{\sigma_f^2}, \quad (3)$$

leads to extremely high values, and, thus, the feeder link is considered noiseless.

With this, the system model in (1) can be simplified as

$$\mathbf{y}^{[i]} = \mathbf{H}^{[i]} \mathbf{x} + \mathbf{n}^{[i]}, \quad i = 1, \dots, N_u. \quad (4)$$

Curiously, in spite of the signal model in (1) presenting strong similarities with the multiuser MIMO amplify-and-forward relay channel [13], it is evident that after the high feeder SNR assumption, the system model in (4) does not consider the relay processing effect.

Getting more into the detail of $\mathbf{H}^{[i]}$, this matrix is usually assumed to follow

$$\mathbf{H}^{[i]} = \mathbf{F}^{[i]} \circ \bar{\mathbf{H}}^{[i]}, \quad i = 1, \dots, N_u, \quad (5)$$

where $\bar{\mathbf{H}}^{[i]} \in \mathbb{C}^{K \times N}$ is a full column-rank matrix containing the antenna radiation pattern and path loss and $\mathbf{F}^{[i]} \in \mathbb{C}^{K \times N}$ denotes a matrix of random entries modeling the propagation effects. The operator $\circ$ denotes the Hadamard product. Despite the similarities with the terrestrial multiuser MIMO case,

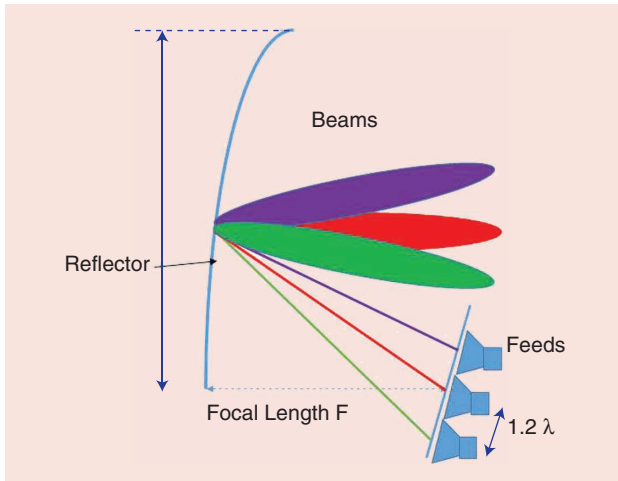


FIGURE 8. An array-fed reflector antenna and the beam-generation process.

both channel models are quite different, as we are going to describe next. The (k, n) th entry of matrix $\bar{\mathbf{H}}^{[i]}$ is

$$[\bar{\mathbf{H}}^{[i]}]_{k,n} = \frac{G_R d_{kn}^{[i]} e^{j\psi_{k,n}^{[i]}}}{4\pi \frac{d_k^{[i]}}{\lambda}} \quad k = 1, \dots, K; n = 1, \dots, N; i = 1, \dots, N_u. \quad (6)$$

with $d_k^{[i]}$ the distance between the i th UT at the k th beam and the satellite, λ the carrier wavelength, and G_R^2 the UT receive antenna gain. The term $d_{kn}^{[i]}$ refers to the gain from the n th feed to the i th user at the k th beam. Because of the high directivity of each feed, these gains are very low for users in beams that are far away from the illuminated area by each feed.

The time-varying phase due to the beam radiation pattern and radio wave propagation is represented by $\psi_{k,n}^{[i]}$. The phase value $\psi_{k,n}^{[i]}$ comprises different contributions. In particular,

$$\psi_{k,n}^{[i]} = \theta_{\text{RF},k}^{[i]} + \theta_{\text{LNB},k}^{[i]} + \theta_{\text{PL},n}, \quad (7)$$

where $\theta_{\text{RF},k}^{[i]} = (2\pi/\lambda)d_k^{[i]}$ is the phase rotation due to the RF signal propagation, which depends on the UT distance to the satellite; $\theta_{\text{LNB},k}^{[i]}$ is the phase contribution of the receiver's low-noise block downconverters, assumed to be Gaussian, with zero mean and a standard deviation of 0.24° ; and $\theta_{\text{PL},n}$ represents the payload oscillator phase offsets that are assumed to be Gaussian, with zero mean and a standard deviation of around 2° , with ultrastable oscillators [14].

We note that the values of $d_{kn}^{[i]}$ depend on the onboard multiantenna solution that is employed. If the satellite uses direct radiating elements (i.e., a phased-array aperture), then $d_{kn}^{[i]}$, with $n = 1 \dots N$, is the steering vector of user k , thus conferring a Vandermonde structure to matrix $\bar{\mathbf{H}}^{[i]}$ whenever the phase $\psi_{k,n}^{[i]}$ is compensated. For instance, in L band satellite systems, the onboard beamforming is done via direct radiating elements (this is the case with the Inmarsat and Globalstar systems). On the contrary, at the Ku and Ka bands, the satellite generally employs a reflector antenna with multiple feeds (i.e., array-fed reflectors), as the one depicted in Figure 8, which presents the required satellite gains at these millimeter-wave frequencies.

Also, in (4), we consider that all feed signals impinge on the UT antenna at the same time instant (i.e., there is no time misalignment between simultaneously transmitted frames). This aspect, together with the assumption of perfect channel state information at the transmitter, enables the precoding techniques that are proposed in next section. If perfect channel state information is not available, the same strategies as in terrestrial wireless communications must be designed. In this case, the satellite scenario does not raise any new signal processing problem, and, therefore, the approaches to address it are not covered in this article.

To finalize the description of (5), we note that the atmospheric fading coefficients from one user to the different antenna elements are the same. This is caused by the relatively small separation of the antenna elements compared with the satellite altitude. Therefore, $[\mathbf{F}^{[i]}]_{k,n} = \mu_k^{[i]} e^{j\theta_k^{[i]}}$, where each fading

coefficient is independent of the transmission feed. Notably, there is no multipath, and a strong line of sight is present in frequencies above 10 GHz (i.e., above the Ku band). Whenever there is no blockage, $\mu_k^{[i]}$ can be assumed to be Ricean distributed; therefore, (4) models a line-of-sight channel.

Precoding techniques

To mitigate the cochannel interference due to the high frequency-reuse factor, precoding is performed, and the transmitted signal vector per beam is given by $\mathbf{x} = \mathbf{W}\mathbf{s}$, where $\mathbf{s} \in \mathbb{C}^{K \times 1}$ is the vector that contains the transmitted symbols per UT, which we assume are uncorrelated and with power equal to one ($E[\mathbf{s}\mathbf{s}^H] = \mathbf{I}_K$). Matrix $\mathbf{W} \in \mathbb{C}^{N \times K}$ is the linear precoding matrix to be designed. As mentioned previously, each frame contains information intended for multiple users to attain a large channel-coding gain. In this context, every UT with index $i = 1, \dots, N_u$ at the k th beam detects the same information $[\mathbf{s}]_k$, leading to the so-called multigroup multicast transmission [15]. While most of the works on multigroup multicast precoding have considered only sum-power constraints at transmission, in SatCom, a per-antenna power constraint needs to be taken into account because of the intrinsic per-antenna power limitation on the satellite.

An important figure of merit to design the system is the system sum rate, which is the aggregate information rate of all of the beams. Specifically, it is defined as $\mathcal{SR} = \sum_{k=1}^K \min_{i=1, \dots, N_u} \log_2(1 + \text{SINR}_k^{[i]})$, where $\text{SINR}_k^{[i]}$ is the SINR of the i th user at the k th beam. This is defined as the ratio between the power of the desired received signal and the power of the noise and interference. It reads as

$$\text{SINR}_k^{[i]} = \frac{|\mathbf{h}_k^{[i],H} \mathbf{w}_k|^2}{\sum_{j \neq k} |\mathbf{h}_k^{[i],H} \mathbf{w}_j|^2 + \sigma_n^2}, \quad (8)$$

where $\mathbf{w}_k$ is the k th column of $\mathbf{W}$. Note that, since we are considering a multicast transmission, the achievable data rate at each beam is determined by the data rate that the UT with the lowest SINR can achieve. In this way, the selected MODCOD for transmission is decodable by all UTs in the frame. The MODCOD selection can be made in practice thanks to the feedback that each user makes of its SINR, as is indicated in reference [12].

As a matter of fact, the solution to the following optimization problem

$$\begin{aligned} \mathcal{P}_1: \quad & \underset{\mathbf{W}}{\text{maximize}} \quad \mathcal{SR} \\ & \text{subject to} \\ & [\mathbf{W}\mathbf{W}^H]_{nn} \leq P \quad n = 1, \dots, N \end{aligned} \quad (9)$$

provides the maximum sum rate that a multicast system can attain. We note that the objective function in problem $\mathcal{P}_1$ is nonconvex. Also, in $\mathcal{P}_1$, a matrix of around 10,000 complex elements should be optimized over hundreds of per-antenna power constraints. The work in [16] considers the optimization of $\mathcal{P}_1$ via an alternating projection technique with a semidefinite relaxation (SDR) procedure, which is adequate for small- to medium-coverage areas (i.e., a small to medium number of

beams). If a notably larger number of beams and/or users is targeted, current nonconvex optimization alternatives might fail because of the immense computational complexity, which opens potential avenues for future research. This is the case with the optimization framework of [17], which tackles nonconvex objective functions over nonconvex constraints.

It is important to remark that, before tackling any precoder design, the scheduling process needs to be considered, as it plays a key role in obtaining relevant sum-rate values in multicast precoding (see [16] and [18]). It is crucial to select the most convenient users to be served in each satellite frame. This scheduling process could simply consider the geographical position of each UT. In this way, information from UTs that are geographically close can be embedded into the same frame to yield efficient data rates. Note that geocustering would not work in a terrestrial context, since geographical proximity does not imply similarity in channels.

On the other hand, the GW can rely on the user channel vector and group users with similar vectors. This is the case with the minimum Euclidean norm scheduling technique (minEuclidean) presented in [12]. This technique randomly selects one user at each beam and afterward finds $N_u - 1$ users whose channel vector has the lowest Euclidean norm with respect to the randomly selected one. Although suboptimal, note that this scheduling process is done distributively for each beam, with an affordable complexity. Further techniques for grouping users to be served over the same frame constitute an unexplored field of investigation. Using a graph to represent the UTs covered by the satellite, with their different features (e.g., channel quality and traffic), allows one to use spectral clustering as a possible alternative, which may render better performance when the optimal clusters are not convex [19].

To alleviate the complexity of $\mathcal{P}_1$, suboptimal precoding alternatives with lower complexity have been reported in the literature. One of the most adopted techniques, which returns a good performance-versus-computational-complexity tradeoff, is the spatial minimum mean square error (MMSE) precoder, which can be formulated for unit variance noise as $\mathbf{W}_{\text{MMSE}} = \beta_{\text{MMSE}} \left(\hat{\mathbf{H}}^H \hat{\mathbf{H}} + \frac{1}{P} \mathbf{I}_N \right)^{-1} \hat{\mathbf{H}}^H$, where β_{MMSE} controls the transmit power to fulfill the per-feed power constraints and $\hat{\mathbf{H}} = (1/N_u) \sum_{i=1}^{N_u} \mathbf{H}^{[i]}$. In other words, this design consists of the MMSE precoding over the average channel matrix of all users simultaneously served at each beam. This is one important peculiarity of the multicast problem when designing a low-complexity precoder, i.e., to discover the best channel metric that represents N_u channels. Averaging is not suitable for all wireless channels, but it works well for the satellite channel, which has a strong line of sight and no phase variation across the user channel vector. This precoder is the so-called UpConst Multicast MMSE. A comprehensive study of different linear precoding techniques for the general multigroup multicast problem can be found in [20]. As an example, Figure 9 shows the average beam data rate, defined as

$$\mathcal{SR} \frac{B}{K}, \quad (10)$$

where B is the user bandwidth. The figure depicts the performance of both the UpConst Multicast MMSE and the block

singular-value decomposition (block-SVD) technique presented in [11]. The block-SVD method was inspired by the seminal work in multiuser MIMO precoding of [21]. In particular, this approach aims at mitigating the interbeam interference by employing the subspace null projection.

To obtain the results, we consider a beam pattern with 245 beams and a maximum per-antenna power constraint of 55 W. The results were obtained over 1,000 Monte Carlo runs and unit variance noise. Figure 9 also depicts the average beam data rate whenever the GW performs a certain scheduling or not (i.e., random scheduling). In particular, we opt to use the minEuclidean scheduling mechanism.

Clearly, the larger the N_u , the lower the attainable rates obtained by both block-SVD and UpConst Multicast MMSE. In all cases, block-SVD leads to larger data rates compared to the UpConst Multicast MMSE. In both cases, the impact of scheduling is remarkable as a key enabling component for yielding a capacity increase with respect to the benchmark case (i.e., the four-color scenario depicted by the black line in Figure 9).

Nevertheless, the computational complexity of UpConst Multicast MMSE is much lower than block-SVD and does not grow notably when the number of users N_u per frame increases. On the contrary, block-SVD requires more computational time to compute the precoding matrix as the number of UTs grows. In any case, despite its low computational complexity, UpConst Multicast MMSE still presents implementation challenges when serving large coverage areas (i.e., the computation of the matrix inverse becomes a computationally demanding operation as K grows). Consequently, the study of alternative precoding designs is of extraordinary interest for both academia and industry. These alternative designs should require only a limited number of operations when computing their precoding matrices, while providing high data rates.

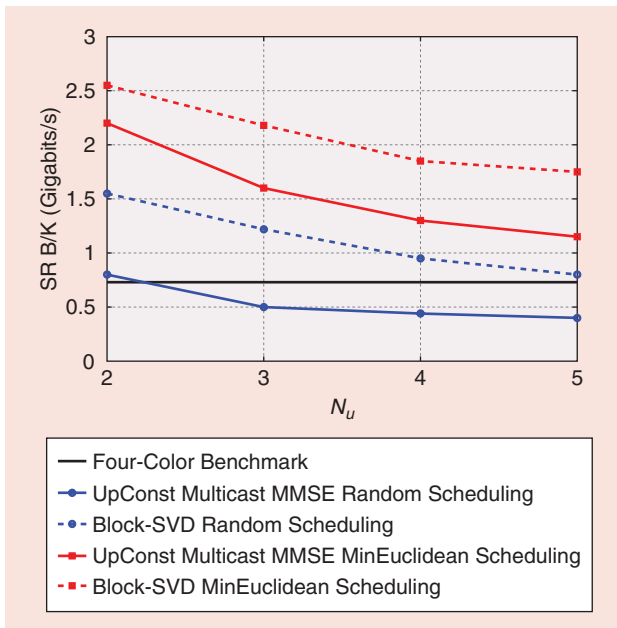


FIGURE 9. The average beam data rate of two precoding techniques, UpConst Multicast MMSE and block-SVD.

Recently, the authors in [22] proposed a linear low-complexity multicast precoding alternative for the massive MIMO context based on a cascaded precoding design with a certain pilot allocation optimization. However, the presented work in [22] cannot be directly mimicked in the multibeam satellite context, since the proposed mechanism assumes a Rayleigh channel distribution and takes advantage of the presence of pilot contamination, which does not occur in SatCom.

Whenever higher layers are considered, the precoding design should be able to guarantee a certain QoS to the UTs. In contrast to cellular systems, satellite operators offer their clients service-level agreements (SLAs) that involve a minimum data rate over a certain percentage of channel access attempts. In this case, the fulfillment of the SLA contracts by precoding is done by optimizing the following problem:

$$\begin{aligned} \mathcal{P}_2: \quad & \underset{\mathbf{W}}{\text{minimize}} \quad \|\mathbf{W}\|^2 \\ & \text{subject to} \\ & [\mathbf{W}\mathbf{W}^H]_{mm} \leq P \quad n=1, \dots, N, \\ & \text{SINR}_k^{[i]} > \gamma_k \quad k=1, \dots, K \quad i=1, \dots, N_u. \end{aligned}$$

The optimization problem $\mathcal{P}_2$ is a nonconvex quadratically constrained quadratic problem (QCQP), which limits its applicability in large-scale coverage areas. This problem can be tackled via SDR approximation methods. Bearing this in mind, efficient parallel implementation of the nonconvex QCQP optimization tools can be a good alternative for solving $\mathcal{P}_2$ in real multibeam satellite systems. This is the case for the work simultaneously done in [23] and [24], which promotes the use of the alternating direction method of multipliers to solve the nonconvex QCQP in $\mathcal{P}_2$. Still, it is an open topic for research to discover good low-complexity alternatives.

The increase in the capacity of the user link requires a corresponding increase in the capacity of the feeder link. In principle, the exploitation of higher-frequency bands by the wireless link could address this issue. However, this approach is often not feasible in practice. In the following, we propose different system architectures able to tackle the feeder link bandwidth limitation aspects.

Multiple GWs

Multibeam precoding over multiple GWs consists of transmitting the precoding signals over geographically separated GWs that are usually interconnected. In this way, the equivalent feeder link can aggregate the bandwidth of the feeder links of the different GWs and can accommodate the bandwidth increase that is needed when frequency reuse increases. Furthermore, in case one of the GWs fails or has very adverse fading, the traffic can be rerouted to maintain the system capacity. The configuration is depicted in Figure 10.

Again, one can find a strong relationship between the multiple-GW multibeam scenario and a multicell C-RAN system, where each cell fronthaul has a limited bandwidth capacity, as does each GW. Remarkably, challenges in both systems are still to be tackled, as we describe in the following. In contrast to the single-GW scheme, multiple-GW precoding presents two

main issues. First, the original precoding matrix $\mathbf{W}$ becomes block-diagonal, such that

$$\mathbf{W} = \text{block-diag}\{\mathbf{W}_1, \dots, \mathbf{W}_l, \dots, \mathbf{W}_L\}, \quad (11)$$

where $\mathbf{W}_l \in \mathbb{C}^{K_l \times N_l}$ is the precoding matrix associated with the l th GW ($l = 1, \dots, L$). Note that, for multiple-GW precoding, $N = \sum_{l=1}^L N_l$, and $K = \sum_{l=1}^L K_l$. In other words, each GW can use only a subset of the N feed signals for performing the interference mitigation. This fact limits the overall system performance, as it reduces the available degrees of freedom. On the other hand, each of the GW feeder link bandwidth requirements is reduced. Indeed, the l th GW transmits only $K_l N_l$ precoded signals instead of the KN signals that were transmitted in the single-GW scenario.

The second main challenge is the channel state information acquisition. Each GW can access only the feedback information from its served users, but each GW needs the channel state information of the adjacent beams to reduce the generated interference. Therefore, a set of matrices must be exchanged by the different GWs, leading to a large communication overhead [25], [26]. This exchange of channel state information is the same sharing challenge as in the terrestrial multicell scenario.

Perfect connectivity between GWs might not be possible in real deployments. In this context, the multiagent optimization of $\{\mathbf{W}_l\}_{l=1}^L$ may be of interest to implement, assuming certain QoS requirements between the different GW connections. This impacts not only on the tentative optimization but also on the design of the compression algorithms for exchanging information from the different GWs. Finally, the precoding structure in (11) is similar to the group sparse beamforming presented in [27]. In light of this, promoting group sparsity in both $\mathcal{P}_1$ and $\mathcal{P}_2$ might result in an efficient multiple-GW precoding design.

Hybrid on-ground/onboard precoding

Another alternative to reduce the bandwidth requirements in the feeder link is to carry out a certain part of the precoding onboard the satellite. This architecture is termed *hybrid on ground/onboard*, and it offers a substantial feeder link bandwidth reduction [28].

Precisely, the original true on-ground precoding matrix is transformed to

$$\mathbf{W} \rightarrow \mathbf{U}\mathbf{V}, \quad (12)$$

where $\mathbf{U} \in \mathbb{C}^{N \times K}$ is the beamforming operation done at the satellite and $\mathbf{V} \in \mathbb{C}^{K \times K}$ is the on-ground beamforming implemented at the GW. In this context, instead of the original precoded signal $\mathbf{W}\mathbf{s}$ of NK user signals, with this hybrid onboard/on-ground architecture, $\mathbf{V}\mathbf{s}$ should be transmitted instead, and it is formed by K^2 user signals. Bearing this in mind, the feeder link band-

width requirement is reduced by a factor of N/K , which can be substantially high in multiple-feed-per-beam architectures. The price to pay for this data compression is a reduction in the final achieved throughput. As [29] studies, this reduction can be up to 20%, and it is because the interference mitigation process acts on the K user beams directly instead of on all of the N radiating elements, thus reducing the available degrees of freedom for those cases for which the number of radiating elements is higher than the number of user beams.

In any case, we note that there are several design choices regarding the partitioning of the processing load between the ground and space segments. Thus, the hybrid onboard/on-ground precoding architecture presents additional challenges apart from the ones introduced in the previous section. Specifically, the onboard beamforming network does not have the same processing capabilities as the one that is purely implemented on the ground. Indeed, current payloads allow beamforming reconfigurability on a monthly or daily basis. This restriction translates into a difference in the transmitter channel state information, i.e., while $\mathbf{V}$ can rely on instantaneous channel state information, the onboard $\mathbf{U}$ should depend on a forecast ergodic value of the channel matrix. Specifically, in [30] the authors study the capacity loss that comes from the fact that the onboard beamforming is fixed or can be updated only in a very slow fashion.

Notably, the proposed scheme in (12) presents certain similarities to current works in millimeter-wave precoding techniques, where hybrid digital–analog beamforming is under study to reduce the complexity that emerges when working with large apertures (i.e., with a high number of elements). These large apertures are needed to achieve high gains at the millimeter-wave range. Remarkably, SatCom has been dealing with large-scale apertures at the millimeter-wave level for a long time, thus developing low-complexity architectures and implementations that are able to manage them. That is why the concept of beamspace communications (i.e., $\mathbf{U}$ is then the so-called beamspace matrix) arose. The beamspace is implemented with analog technology on the satellite, while the digital part

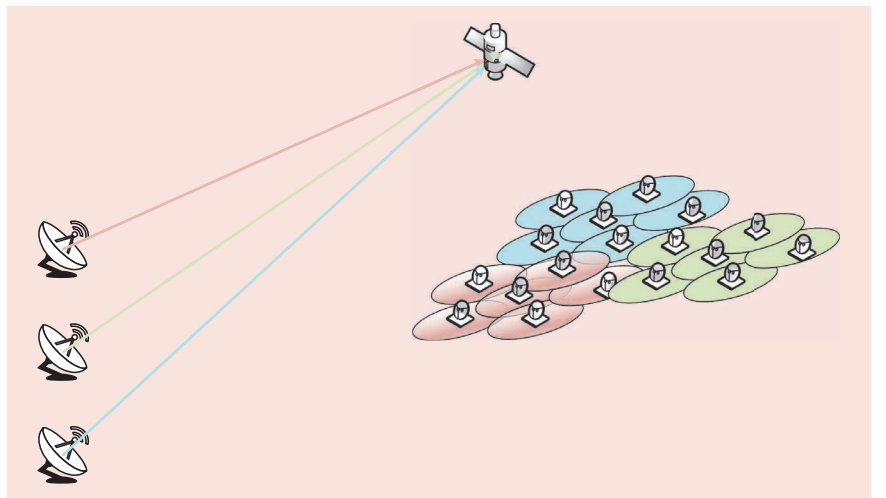


FIGURE 10. A multi-GW architecture configuration.

of the precoder is implemented at the GW (see [31] and [32] for more details). In fact, beamspace communications are now being reconsidered in millimeter wave, but with phased-array technology (e.g., see [33]) instead of the usual array-fed reflector in SatCom.

The inspiring recent work of [34] introduces the idea that $\mathbf{U}$ is a sparse matrix. Indeed, considering large coverage areas, beams that are sufficiently separated do not interfere with each other. Therefore, onboard beamforming can reduce implementation costs, and the feeder link can limit the number of transmitted precoded signals, if $\mathbf{U}$ is designed to be sparse. This idea can be introduced for the multicast scenario in $\mathcal{P}_2$ by separately incorporating the matrices $\mathbf{U}$ and $\mathbf{V}$ in the optimization. This is a challenging problem because of the nonconvex constraints.

Finally, recent preliminary investigations show that the mass and cost of a satellite payload can be greatly reduced if the onboard beam generation is done via sparse arrays. This restricts the $\mathbf{U}$ to having zero or one entries, resulting in a most challenging variation of the $\mathcal{P}_2$ optimization problem.

Summary of precoding in multibeam satellite systems

The following list summarizes the main peculiarities of precoding in multibeam satellite systems:

- The antenna systems have extremely large-scale dimensions.
- There is multicast transmission.
- Per-antenna power constraints instead of sum-power constraint have to be considered.
- More than one GW may serve one satellite to alleviate the capacity limitations of the feeder link.
- The satellite system architecture and complexity impose constraints that are different than the ones in the terrestrial segment, motivating hybrid onboard/on-ground precoding architectures.

Finally, Table 1 summarizes the aspects that we have considered in this section as well as their current signal processing challenges. The next section deals with signal processing on board the satellite, which introduces additional degrees of processing and performance improvement when compared to the traditional satellite approach that uses a transparent payload.

Onboard signal processing

H/VHTSs do not offer the possibility of hardware replacement once launched into orbit. This motivated the traditional payload designs where the minimum necessary processing was performed on board the satellite. Furthermore, such processing was performed in the analog domain because of its techno-

logical maturity in supporting high bandwidth while sustaining satellite platform constraints, including power limitations, heat dissipation, and radiation. This resulted in the transparent payload architecture, where the satellite is a passive relay, performing only frequency translations (a feeder-to-user link and vice versa), channelization, and amplification.

With improvements in space-hardened digital circuitry, a few satellites with digital OBP are being designed [35]. In these spacecraft, the OBP is used to enable digital synthesis of narrow-band user/beam specific carriers from the incoming wideband stream using filter banks. OBP further enables programmable routing of such carriers to end users. The aforementioned OBP functionalities suffice for most of the current transparent satellites based on link budget design. However, with the gradual design shift from the link budget-based to the interference-limited paradigm, there is a need for considering advanced digital signal processing/communication algorithms for interference management. Similar to the advantages offered by relay processing in terrestrial networks, OBP offers additional degrees of freedom for efficiently implementing interference management algorithms. This motivates the discussion on OBP pursued in this section.

State of the art in OBP

Providing limited digital processing on board the satellite is not a new concept and has been discussed in the last decades [36]. The key OBP paradigms observed from these developments can be categorized as follows:

- *Regenerative processing*: This is the straightforward way to OBP. It involves generating the digital baseband data on board after waveform digitization, demodulation, and decoding. This is similar to the decode-and-forward paradigm in relay systems and is considered for multiplexing different streams, switching, and routing [1]. Clearly, regenerative processing provides better noise reduction and flexibility. However, its complexity is rather large for H/VHTSs because of the high bandwidths used. Needless to say, such processing needs to be reconfigurable to accommodate evolutions in the physical layer.
- *Digital transparent processing (DTP)*: This is a simpler approach to OBP, operating only on the samples of the input waveform. The amplify-and-forward architecture in relaying is a simple DTP. Since neither demodulation nor decoding is implemented [36], DTP processing results in payloads that are agnostic to air-interface evolutions. Typical applications include digital beamforming,

Table 1. The benefits and signal processing challenges of different multibeam satellite precoding functionality.

System Architecture	Benefits	Signal Processing Challenges	References
Single-GW precoding	System throughput increase	User clustering, large-scale precoding optimization	[11], [16]
Multiple-GW precoding	Per-GW feeder link bandwidth reduction, GW redundancy (outage reduction)	Limited channel state information, GW sharing, large-scale distributed optimization, precoding under per-feeder link bandwidth constraints	[26], [25]
Precoding with onboard beamforming	Feeder link bandwidth reduction, onboard antenna cost reduction	Onboard beamforming (beamspace MIMO) design with payload antenna restrictions	[30], [34]

broadcasting/multicasting based on single-channel copies, RF sensing, and path calibration [36].

- **Hybrid processing:** This is an interesting paradigm that involves digitizing the entire waveform but regenerating only a part for exploitation. As a case in point, the header packet is regenerated to allow for onboard routing [32].

New drivers for OBP: Signal processing algorithms

As discussed in the section “High Throughput Satellites: A New Interference-Limited Paradigm,” satellite systems are evolving from link budget-limited to interference-limited designs. Advanced signal processing like interference mitigation for such evolving scenarios is being accommodated currently through on-ground implementation at the GW. However, OBP provides additional degrees of freedom that can be exploited to yield an implementation of these signal processing algorithms more efficiently than those implemented on the ground. The various advantages offered by relays with processing capabilities, in terms of diversity, rate, and so forth, are well explored in the terrestrial literature. However, OBP in SatCom offers the following particular features that necessitate novel signal processing applications:

- **Latency reduction:** Because of long round-trip delays between the terminals on the ground (terminal–satellite–terminal), there is a large latency (250 ms) before the effect of on-ground processing at one communication terminal is discovered at another. The delay can be reduced by half through OBP, thereby enhancing the efficiency of the underlying techniques. This feature is particularly useful for precoding to mobile terminals where round-trip delay severely affects the fidelity of the channel state information at the transmitter; onboard precoding can reduce the degradation due to perturbed channel state information.
- **Information accessibility:** Since the satellite aggregates information from multiple GWs or UTs before appropriate channelization, it has more information than the individual GWs or UTs. This enables joint processing on board, without the additional cost of sharing information across GWs/UTs, and is particularly useful in multiple-GW scenarios where inter-GW interference can be easily mitigated on board.
- **Support for evolved and emerging techniques:** Since many of the challenges and constraints arise on board the satellite, OBP possesses the wherewithal to address them. OBP extends support to such pre-distortion and emerging techniques as full-duplex operations and anti-jamming. For example, full-duplex relaying by satellite requires canceling self-interference on board the satellite in both the analog and digital domains.

The OBP methods used thus far have focused on networking, such as onboard switching, traffic routing, and multiplexing data/multimedia [1], with limited signal processing applications per se. However, novel signal processing techniques, as those in Table 1, would greatly benefit from the additional degrees of freedom offered by OBP. Hence, the emerging signal processing algorithms serve as drivers for further proliferation of OBP. The benefits of OBP are illustrated next through a simple signal processing application in interference detection.

Interference detection: Exploiting different flavors of OBP

As an illustrative example, we consider the detection of interference at the satellite generated from on-ground terminals, either maliciously or because of improper installation. The scenario is depicted in Figure 11. Such an intersystem interference appears on the uplink; however, these unwanted transmissions corrupt the desired signal being relayed, thereby reducing the end-user SINR and significantly impacting the operations.

Currently, such uplink RF interference is detected and localized on the ground from a perturbed downlink transmission. Subsequently, the interference is mitigated by satellite operators using standard manual procedures. However, this methodology suffers from longer reaction times and performance degradation. The latter arises from the fact that additional noise (receiver front end, propagation, and such) and distortions from the satellite transponder corrupt the downlink signal that is used for on-ground detection [37].

On the other hand, onboard interference detection and localization can be undertaken by introducing a dedicated spectrum-monitoring unit in the satellite payload that can take advantage of the emergent OBP capabilities. This provides for a faster reaction time and enhances detection capability, since it avoids additional downlink noise and payload-induced perturbations. For the sake of exposition, we consider the detection exercise

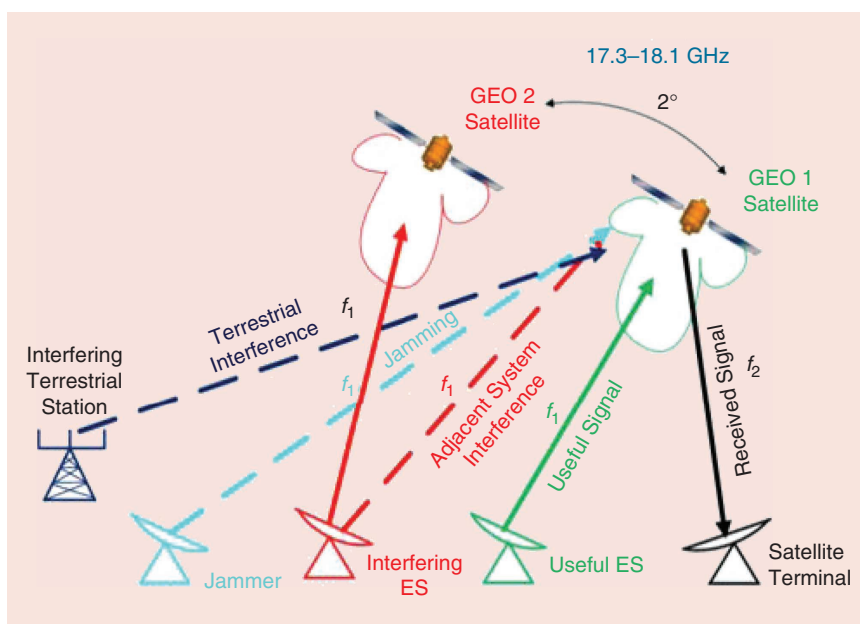


FIGURE 11. An illustration of interference affecting satellites. ES: Earth Station.

herein and illustrate how the degrees of freedom offered by OBP can be leveraged.

We examine a generic scenario in which the satellite, the desired GW, and the interferer are each equipped with a single antenna. We further assume ideal sampling and digitization on board. Under these conditions, detection of the uplink RF interference can be formulated as the following binary hypothesis-testing problem:

$$\begin{aligned} \mathcal{H}_0: & \tilde{x}_k(n) = h s_k(n) + \eta_k(n), \quad n = 1, 2, \dots, T, \\ \mathcal{H}_1: & \tilde{x}_k(n) = h s_k(n) + \eta_k(n) + p_k(n), \quad n = 1, 2, \dots, T, \end{aligned} \quad (13)$$

where T is the number of samples used for detection, and h denotes the scalar flat-fading channel from the desired GW to the satellite. Furthermore, let $s_k(n)$ be the sample of the intended signal transmitted by the desired GW on the k th channel (or stream) at instance n . Similarly, let $p_k(n)$ be the interfering signal on board and $\eta_k(n)$ the noise receiver modeled as a realization of an independent and identically distributed complex Gaussian variable with zero mean and unit variance.

For such a problem, several interference-detection techniques can be implemented on board.

- **Conventional energy detector (CED):** This is a technique that works on samples $\tilde{x}_k(n)$ directly and chooses a hypothesis based on $\sum_{n=1}^T |\tilde{x}_k(n)|^2 \stackrel{H_1}{\underset{H_0}{\gtrless}} \gamma_1$, where γ_1 is a threshold [38]. Such a detector can be easily implemented with a DTP. While a CED is shown to be effective for strong interference, it is rather susceptible to variations in the noise power.
- **Energy detector with signal cancellation on pilots (EDSCP):** This method exploits the frame structure in the transmitted waveform to estimate the channel h on pilot symbols [the known $s_k(n)$] [39]. Subsequently, the EDSCP detector performs the detection test, $\sum_{n \in \text{Pilots}} |\tilde{x}_k(n) - \hat{h} s_k(n)|^2 \stackrel{H_1}{\underset{H_0}{\gtrless}} \gamma_P$ for some threshold γ_P . Here, $\hat{h}$ is the channel estimate obtained on the pilots through standard processing, and $\tilde{x}_k(n) - \hat{h} s_k(n)$ is an estimate of the interference $p_k(n)$ at the pilot

locations. This is an example of hybrid processing, where only the frame header is decoded to determine the location of the pilots.

- **Energy detector with signal cancellation on data (EDSCD):** Initially proposed in [40] and further developed in [41], this technique first detects $\{s_k(n)\}$ and subsequently removes its contribution from $\tilde{x}_k(n)$. This facilitates an estimation of the interference, not only on pilots but also on data. The EDSCD detector takes the form $\sum_{n=1}^T |\tilde{x}_k(n) - \hat{h} s_k(n)|^2 \stackrel{H_1}{\underset{H_0}{\gtrless}} \gamma_D$ for some threshold γ_D , where $\hat{h}$ is the channel estimate obtained on the pilots and $\{\hat{s}_k(n)\}_{n \notin \text{Pilots}}$ are the detected symbols. EDSCD requires regenerative processing.

Figure 12 presents the probability of detection as a function of the received interference-to-signal-plus-noise ratio (ISNR), comparing the CED, EDSCP, and EDSCD detection schemes. The ISNR is chosen as a performance metric to reflect the fact that interference is the signal of interest in this application, while the other quantities are unwanted. In practice, there is an uncertainty of 0 to 1 dB in the variance of $\eta_k(n)$ in (13); this uncertainty is represented as ϵ in the legend in Figure 12. For this setup, we consider $T = 516$ modulated symbols comprising $T_d = 460$ data symbols and $T_p = 56$ pilots, representing a realistic waveform according to the DVB-RCS2 standard. It is observed that the interference detection performance decreases with uncertainty. The latter may lead to the ISNR wall phenomenon, where the detectors cannot recognize the interference robustly beyond a certain ISNR value. Furthermore, we see that the EDSCP and EDSCD schemes perform considerably better than the CED with uncertainty, improving the ISNR wall by more than 5 dB. It should be noted that all of the algorithms can be implemented on the ground; however, their performance would be penalized by additional payload impairments and downlink noise. Thus, classical interference detection problems can be dealt with via different onboard architectures, with sophisticated processing providing additional performance benefits.

Architecture of an onboard processor

Having reviewed the state of the art and demonstrated the usefulness of OBP with an illustrative example, we now proceed to detail the OBP block of Figure 4. Such a system description will lead to the modeling of key OBP blocks, enabling further understanding of the system and of the design of processing algorithms. The details are given in Figure 13, which presents a payload transponder employing digital OBP. Standard analog front-end receiver processing is carried out prior to the digital processing. This includes filtering, low-noise amplification, mixer, and automatic gain control; these are used in down-converting the input RF signal to an appropriate intermediate frequency (IF). The key OBP components are detailed as follows.

High-speed analog ADC and baseband/IF conversion
Assuming an IF signal with a maximum bandwidth of $2f_c$ centered around f_c , ADCs sample at a frequency of $F_s \geq 4f_c$ to avoid aliasing. Subsequently, the resulting samples are converted to baseband (in- or quadrature-phase channels), using appropriate filtering [42].

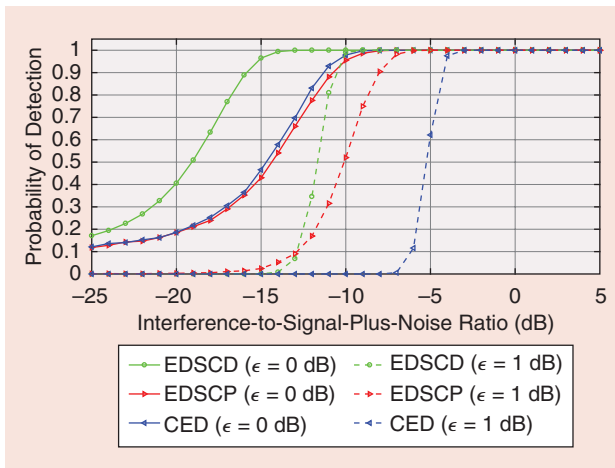


FIGURE 12. The performance of various onboard interference detection schemes [quadrature phase-shift keying modulation for $s_k(n)$, $T = 516$, SNR = 6 dB].

Analysis filter banks

The baseband/IF signal is a multiplex of carriers of varying bandwidth, each carrier broadcast over a beam or addressed to a particular user. A first step toward dedicated processing is to demultiplex the digitized stream into user/beam allocations. Toward this, the baseband/IF input is spectrally decomposed using a filter bank, where the output of each filter corresponds to the smallest quantum of user bandwidth. Typically, non-critically sampled implementation of the analysis filter bank is considered, and a polyphase structure is used. With reference to Figure 13, the input signal is decomposed into S sub-bands using filters $\{H_k\}_{k=1}^S$, each having the smallest quantum of user bandwidth, and the sampling factor is D , with $D \leq S$. Furthermore, the filters are modulated versions of a primary filter, leading to a fast Fourier transform-based polyphase matrix implementation.

Digital processing block

This generic block subsumes both transparent and regenerative architectures. Since the bandwidth allocation to users/beams varies, the processing block is designed to work on the smallest carrier bandwidth used. It implements the processing of individual streams (e.g., blocks P_i , Q_i), including predistortion, modulation/demodulation, and encoding/decoding. The MIMO blocks impart a joint processing capability among blocks P_i , Q_i , e.g., precoding. In the case of a transparent architecture, these blocks can implement waveform manipulation techniques on one or more outputs of the filter bank; typical examples include a lookup table (LUT) for predistortion, beamforming, precoding, and spectral analysis. In addition, regenerative processing mandates operations defined by the air-interface protocol for generating the information bits and the baseband signal.

Switching

The outputs of all transparent/regenerative processing chains are input to a switch matrix that effects routing in the spatial domain (e.g., from one beam to another), temporal domain (e.g., store and forward), and spectral domain (e.g., frequency hopping). The switching block is implemented through controlled memory reads and writes.

Synthesis filter bank and DAC

A key component of the payload is the HPA, which typically amplifies RF signals from one or two beams. Toward generating the HPA input signals, the synthesis filter bank and DAC implement the process of converting the digital samples in baseband to an IF and finally to the RF domain. Such an implementation is similar to the ADC and analysis filter bank counterparts. With reference to Figure 13, the synthesis filter comprises S filters $\{F_k\}_{k=1}^S$ and an Allocation block, which multiplexes relevant outputs to cater to the case when beams/users are allocated an integer multiple of the smallest quantum of bandwidth. Note that the filter banks without additional processing are designed to be near-perfect reconstructing pairs [43].

Critical to the implementation and operation of onboard processors is the accuracy of the processing chain. Because of the link budget constraints of satellite systems, mildly inaccurate processing chains can severely impact availability and performance. Hence, it is imperative to thoroughly study various imperfections induced by the digital processing and incorporate them in the design of onboard techniques. The nonidealities are listed as follows (additional details can be obtained from [44]):

- quantization errors induced by the ADC conversion
- nonidealities in the filter implementation and use of fixed-point operations
- impairments due to phase noise and carrier offsets.

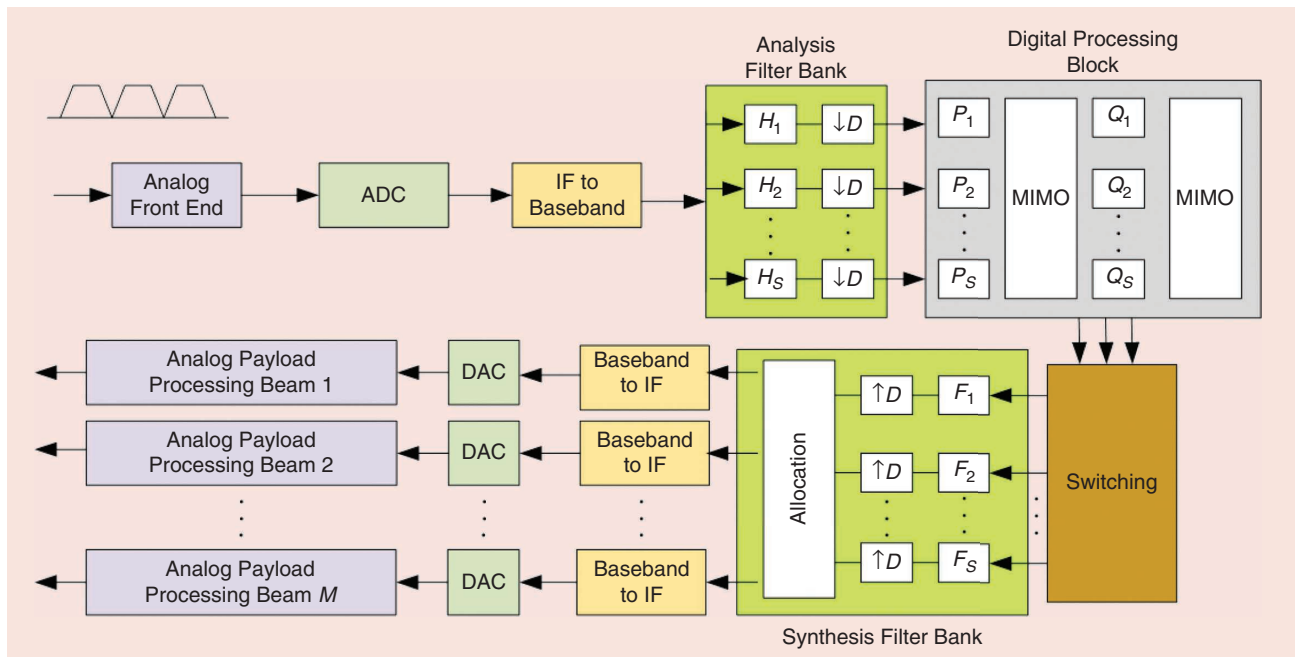


FIGURE 13. The generic architecture of an onboard processor.

Signal model and OBP design challenges

For ease of comprehension, we focus here on a DTP, since the modeling of regenerative payloads is rather intractable. Let $x_k(t)$ be the analog signal corresponding to the k th frequency subband after the analysis bank. The signal $x_k(t)$ can be routed to any beam after appropriate switching. Assuming ideal filtering (i.e., the rejection of out-of-band interference and adjacent channel interference, which would otherwise appear as in-band noise when using practical filters), a DTP would provide the designer access to the samples $\tilde{x}_k(n)$ (at the input of the digital processing block in Figure 13) where

$$\tilde{x}_k(n) \approx e^{j(2\pi[\Delta f]n + \theta + \omega_{n,k})} x_k(nT_s + \tau_{n,k}) + \eta_k(n), \quad (14)$$

where T_s is the effective sampling rate (after the downsampling); $[\Delta f]$ and θ are the frequency and phase offsets, respectively, of the carrier, assumed to be independent of k ; $\tau_{n,k}$ is the sampling jitter; $\omega_{n,k}$ considers the phase noise; and $\eta_k(\cdot)$ represents the noise before the processing block. It should be noted that $\eta_k(n)$ possesses a flat spectral density over the subband k only when the analysis filters are ideal. In addition, Doppler frequencies exist, depending on the orbit; these can be included in Δf , as appropriate.

Let $g_{D,k}(\cdot)$ be the functional equivalent of the OBP (digital processing and filtering) for the k th stream. To include the joint processing and intersymbol interference introduced by filtering, $g_{D,k}(\cdot)$ can be typically modeled as a multiple-input, single-output function with memory. In particular, $g_{D,k}(\cdot)$ can take samples from different streams and time instances as input, i.e., $\{\tilde{x}_l(m)\}_{l,m}$. Furthermore, let $g_{P,l}(\cdot)$ denote the equivalent transfer function of the analog payload processing block for the l th payload. Note that, because of switching and allocation, the k th digital stream is processed by the π_k th payload. In this article, we assume that $\pi_k = k$. Hence, the output of the k th subband analog payload processing takes the form

$$y_k(t) = g_{P,k}\left(g_{D,k}(\{\tilde{x}_l(m)\}_{l,m}) + \widehat{\eta}_k(t)\right) + \widetilde{\eta}_k(t), \quad (15)$$

where $\widehat{\eta}_k(t)$ is the perturbation arising from the noise contributions from the synthesis filter banks and DAC. Furthermore, $\widetilde{\eta}_k(t)$ refers to the noise in the analog processing part of the k th subband. The presence of $\widetilde{\eta}_k(t)$ in (15) caters to the situation where the feeder link is nonideal.

Because of the high carrier frequency, large bandwidths, and adverse operating conditions, the perturbations introduced by the processing chain are significantly large and highly varying compared to those encountered in current terrestrial systems. This, coupled with the limited link margins used in satellites, necessitates that any OBP algorithm operating on the digitized samples must take cognizance of 1) perturbations in the input waveform, as presented in (14), and 2) the impact of processing on noise at the output of the payload, as presented in (15).

In addition, because of the limited power on board, the OBP algorithms need to have low-complexity, efficient implementations. These warrant modification of existing on-ground algorithms and make OBP algorithm design and implementation

challenging. In the ensuing section, a low-complexity OBP algorithm design to minimize impairments will be discussed.

Impairment-cognizant onboard predistortion using DTP

As an illustrative example, we consider the implementation of predistortion on board the satellite. This is motivated by the significant effort expended in the SatCom community on predistortion techniques toward countering the nonlinear impairments introduced by the onboard HPA and achieving higher power and spectral efficiencies [45].

Signal predistortion (SPD), where the waveform is nonlinearly transformed to mitigate the HPA nonlinearities, has been considered in the literature (see the references in [46] for a detailed list). These techniques have been traditionally implemented on the ground. By virtue of being a nonlinear operation, SPD results in a bandwidth expansion; when implemented on the ground, this can have serious consequences in SatCom systems, where transmitted signals must satisfy tight emission masks. Alternatively, additional constraints need to be imposed on SPD to satisfy the mask, which leads to diminished gains. However, such limitations do not appear in onboard SPD because of its colocation with the HPA. Moreover, the LUT approach provides for an attractive alternative for implementation on board rather than functional evaluation.

Modeling onboard SPD

We consider a simple payload model, where the analog processing prior to DTP is denoted by an input multiplexing (IMUX) filter used to remove out-of-band signals from the signal being processed. Furthermore, the analog processing subsequent to the DTP comprises an HPA and an output multiplexing (OMUX) filter for mitigating out-of-band emissions and naturally serves to regulate the output spectrum. With reference to the DTP architecture in Figure 13, the block P_k implements SPD for the k th stream. Focusing on a generic k th stream and omitting the filter bank for ease of analysis, the resulting payload architecture can be simplified to the schematic shown in Figure 14.

In this setup, let $x_k(t)$ be the analog signal at the input of the ADC. An ideal ADC would perform the digitization, giving as output the sampled signal $x_k(n) = x_k(n/F_s)$, where F_s is the sampling frequency. However, this is seldom the case in practice; high-speed ADCs suffer from clock jitter caused by phase noise affecting the clock oscillator. The jitter generates a nonuniform sampling of the input signal, leading to a degradation of the SNR inside the transponder [47]. Under some mild assumptions [48], the SPD input can be described as

$$\tilde{x}_k(n) \cong x_k(n) + e_k(n)\dot{x}_k(n) + \eta_k(n), \quad (16)$$

where $\dot{x}_k(n)$ is the sampled first-order derivative of $x_k(t)$ [48], and $e_k(n)$ is the jitter error signal. The quantization noise and ambient thermal noise are included in $\eta_k(n)$. A typical SPD model involves a third-order memoryless polynomial function taking the form

$$r_k(n) = \gamma_k \tilde{x}_k(n) + \delta_k |\tilde{x}_k(n)|^2 \tilde{x}_k(n). \quad (17)$$

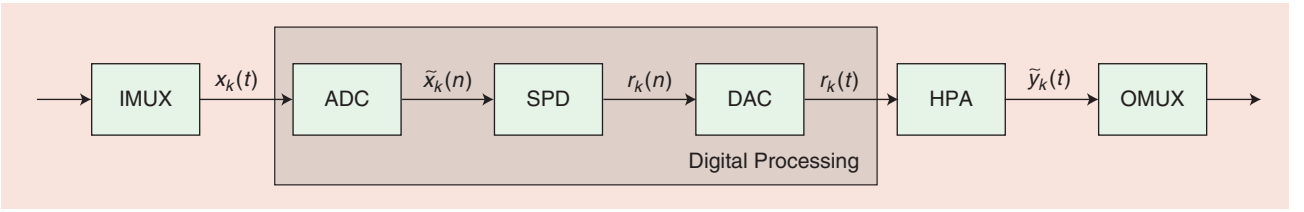


FIGURE 14. A block scheme of the considered transponder with DTP.

On the other hand, the HPA can be modeled as a nonlinear noninvertible memoryless function approximated by a memoryless third-order Volterra series expansion as

$$\tilde{y}_k(t) = \alpha_k r_k(t) + \beta_k |r_k(t)|^2 r_k(t), \quad (18)$$

where $r_k(n)$ is the sampled version of $r_k(t)$ using an appropriate sampling rate. Relating (17) and (18) in the context of (15) (ignoring filter banks), we can immediately recognize that $g_{D,k}(\cdot)$ is a single-input, single-output memoryless function taking the form $g_{D,k}(\tilde{x}_k(n)) = \gamma_k \tilde{x}_k(n) + \delta_k |\tilde{x}_k(n)|^2 \tilde{x}_k(n)$ and $g_{PL,k}(r_k(t)) = \alpha_k r_k(t) + \beta_k |r_k(t)|^2 r_k(t)$. Furthermore, $\hat{\eta}_k(t)$, $\tilde{\eta}_k(t)$ are assumed to be zero. The aim is to determine the implementation gains of the onboard SPD; central to such an implementation is the estimation of the predistorter coefficients minimizing a meaningful metric. This exercise is described next.

Parameter estimation and SPD implementation

As a first step, the HPA model parameters α_k, β_k need to be determined. They are jointly estimated by minimizing $\alpha, \beta = \arg\min_{\alpha, \beta} E_{r_k} \{ |\tilde{y}_k(t) - y_k(t)|^2 \}$, where $y_k(t)$ is the output of the actual HPA and $E_{r_k}\{\cdot\}$ denotes the expectation with respect to the signal $r_k(t)$. This is a least-squares minimization problem, where the error function is linear in the coefficients α_k and β_k . We choose the least-squares cost function since 1) it adequately represents the minimization of modeling errors and 2) it is tractable, leading to an elegant solution without the need for additional information.

The SPD parameters γ_k and δ_k can be found by resorting to a similar minimization problem, i.e., $\{\gamma_k, \delta_k\} = \arg\min_{\gamma_k, \delta_k} E \{ |y_k(t) - x_k(t)|^2 \}$, where $E\{\cdot\}$ denotes the expectation with respect to the signal $x_k(t)$, the jitter $e_k(n)$, and the noise $\eta_1(n)$. Herein, we again choose the least-squares cost function, since it represents the MSE between the transmitted and received symbols and does not require additional information. However, the error function is now nonlinear in the coefficients γ_k and δ_k because of (18), and the method used to estimate the Volterra coefficients of the HPA can no longer be used. Several techniques have been proposed in the literature, including direct learning [46], [49] (based on least mean squares/recursive least squares) and the divide-and-concur algorithm (based on message passing) [50].

While the SPD output $r_k(n)$ can be generated using (17), LUTs provide for a low-complexity solution. Since the dynamics of the input signal $\tilde{x}_k(n)$ are typically known (from calibration tests for the DTP), an LUT catering to the dynamic range can be calculated. Such an LUT implementation follows the memory–performance tradeoff.

Results

Figure 15 compares the performance of the onboard SPD with its on-ground counterpart, where each has a similar structure [50]. The figure also illustrates the performance of onboard SPD optimized with and without the cognizance of jitter. Figure 15 shows that the onboard SPD outperforms its on-ground counterpart both in terms of SINR [providing a 0.4-dB gain at output backoff (OBO), i.e., OBO = 4 dB] and in terms of OBO (providing a 2-dB gain at SINR = 9.5 dB). The on-ground SPD cannot compensate for the IMUX distortions because of its memoryless nature, a shortcoming not encountered by OBP. Figure 15 also shows gains when jitter statistics are accounted for during the optimization of the SPD coefficients. To put the SINR gains in perspective, it should be noted that the DVB-S2X standard allows the use of a MODCOD with higher spectral efficiency [6] for small SINR changes. Furthermore, the OBO gain is of interest to satellite operators because it translates into a power-efficient amplification.

This example shows how OBP provides a platform for signal processing algorithms to exploit, providing gains over on-ground processing. It opens up avenues for further investigation, including studies on sophisticated multistream interference mitigation techniques like precoding.

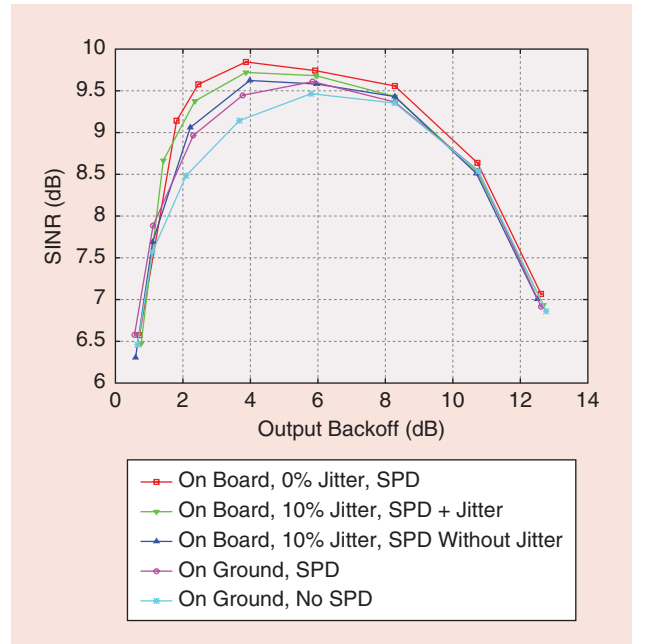


FIGURE 15. The SINR versus output backoff for the jitter-cognizant onboard SPD, compared to its on ground counterpart.

Conclusions and future prospects

H/VHTSs offer far more advantages than their predecessors. This article covered the main signal processing challenges and tools that can be used to boost the spectral efficiency that these satellites can offer. We first discussed high frequency reuse in the multiple beams of these satellites and the role of on-ground processing at the GW. Next, with most of the traffic carried over being digital, signal processing on board the satellite comes as a natural consequence; this overcomes some drawbacks or impairments of on-ground processing.

The focus of the article has been on FSSs. However, Sat-Com refers to a wide range of systems operating in various frequencies and providing different types of services. Future prospects for this work include the application and adaptation of the explained signal processing techniques to mobile satellite services. There are also high expectations in the so-called megaconstellations of MEO and LEO satellites. In these cases, the signal processing concerns are Doppler compensation and intersatellite communications, to mention just two important aspects. Non-GEO satellites require scanning UTs that are more complex than the GEO ones. Finally, note that the feasibility of the proposed solutions depends very much on not only their complexity but their compatibility with legacy systems, full integration with wireless terrestrial communications networks, UT complexity, and final adopted policy.

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Authors

Ana I. Pérez-Neira (aperez@cttc.es) received her B.Sc. degree in electrical engineering from the Universitat Ramon Llull, Spain, in 1989, and her M.Sc. and Ph.D. degrees in telecommunications from the Universitat Politècnica de Catalunya, Spain, in 1991 and 1995, respectively. She has been a full professor at the Universitat Politècnica de Catalunya, Barcelona, Spain, since 2006 and was vice rector for research in 2010. She is a research fellow at the Centre Tecnològic de Telecomunicacions de Catalunya, Barcelona, Spain. Her area of interest is signal processing for communications, focused on satellite communications. She has been an associate editor of *IEEE Transactions on Signal Processing* and *EURASIP Journal on Advances in Signal Processing*. She has been a member of the Signal Processing Theory and Methods (SPTM) Technical Committee (TC) and is currently a member of the IEEE Signal Processing for Communications and Networking TC and of the Board of Governors of the IEEE Signal Processing Society. She was a recipient of the 2018 EURASIP Society Award and is the general chair of IEEE ICASSP 2020.

Miguel Ángel Vázquez (mavazquez@cttc.es) received his B.S., M.S., and Ph.D. (cum laude) degrees in telecommunication engineering from the Universitat Politècnica de Catalunya (UPC), Barcelona, Spain, in 2009, 2012, and 2014, respectively. He also holds a degree in computer science from the National Distance Education University. In parallel with his Ph.D. studies, he has worked as a research engineer at the Centre Tecnològic de Telecomunicacions de Catalunya, Barcelona, Spain, since January 2010. He was a visiting researcher at Patavina Technologies, Padua, Italy, in the context of a Marie Skłodowska-Curie Action. He has played a key role in several projects funded by the European Commission, the European Space Agency, the Spanish Ministry of Science, and various industries.

M.R. Bhavani Shankar (bhavani.shankar@uni.lu) received his M.S. and Ph.D. degrees in electrical communication engineering from the Indian Institute of Science, Bangalore, in 2000 and 2007, respectively. He was a postdoctoral researcher in the Signal Processing Laboratory, KTH Royal Institute of Technology, Stockholm, Sweden, until September 2009. He joined SnT, University of Luxembourg, in October 2009 as a research associate and is currently a research scientist. He is a member of the Executive Committee of the IEEE Benelux Joint Chapter and managing editor of Elsevier's *Signal Processing*. He was a corecipient of the 2014 Distinguished Contributions to Satellite Communications Award from the IEEE Communications Society.

Sina Maleki (sina.malek@gmail.com) received his B.Sc in electrical engineering from Daneshgah Elm va Sanat e Iran in 2006, his M.Sc. degree in electrical engineering and telecommunications from Delft University of Technology, The Netherlands, in 2009, and his Ph.D. degree from the Delft University of Technology in 2013. From July 2008 to April 2009, he was an intern at the Philips Research Center, Eindhoven, The Netherlands, working on spectrum sensing for cognitive radio networks. Since August 2013, he has been with the Interdisciplinary Centre for Security, Reliability, and Trust, University of Luxembourg, where he is focusing on cognitive radio for satellite communications, hybrid satellite-terrestrial backhaul networks, and spectrum monitoring for satellite communications in diverse European Union FP7/H2020 projects, Luxembourgian national projects, and European Space Agency projects.

Symeon Chatzinotas (symeon.chatzinotas@uni.lu) received his M.Eng. degree in telecommunications from the Aristotle University of Thessaloniki, Greece, in 2003 and his M.Sc. and Ph.D. degrees in electronic engineering from the University of Surrey, United Kingdom, in 2006 and 2009, respectively. He is the deputy head of the SIGCOM Research Group, University of Luxembourg. He has authored more than 350 publications, which have received 3,500 citations, and has a Google Scholar h-index of 30. He was a corecipient of the 2014 IEEE Distinguished Contributions to Satellite Communications Award, the CROWNCOM 2015 Best Paper Award, and the 2018 EURASIP Journal on Wireless Communications and Networking Best Paper Award. He is a Senior Member of IEEE.

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